Financial Flexibility with Non-Exclusive Lending

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March 11, 2026

Abstract

This paper examines optimal financial flexibility for firms under financing constraints and non-exclusive lending. We develop a model in which a borrower requires funding for an initial investment and seeks additional financing later following a privately observed liquidity shock. Non-exclusivity creates incentives to dilute initial debt, leading to excessive total borrowing. The optimal contract is an endogenous debt limit: firms with mild shocks retain flexibility but over-borrow, while those with severe shocks face binding constraints and under-borrow. This limit optimally trades off debt dilution against liquidity needs.

Keywords: financial flexibility, debt dilution, lending non-exclusivity, covenant, optimal delegation

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High yield covenants always seek to strike a delicate balance …On the one hand, the covenants provide protection for high yield investors against an issuer's overextending itself or unwisely using cash …On the other hand, the covenants must provide flexibility for the issuer to operate its business and grow …In other words, the covenants protect the investors' ability to be paid principal and interest …while preserving the issuer's ability to run its business and grow without undue restrictions. (Simpson Thacher & Bartlett LLP, 2022)

1 Introduction

Financial flexibility refers to a firm's ability to obtain financing when needed. According to the Duke CFO survey, it is the most important factor in debt and capital structure decisions, and its importance has grown significantly over the past two decades (Graham, 2022). One reason for its importance is that it allows firms to respond effectively to future uncertainties, such as pursuing attractive investment opportunities or dealing with financial distress. Raising external financing can be challenging in such situations because firms may face difficulties in credibly communicating their financing needs to external financiers. Therefore, during financial planning, firms preserve pre-arranged financial flexibility to maintain discretion over their borrowing needs in the future.

While valuable, excessive financial flexibility can be harmful to the firm value. For instance, firms with free cash flows may undertake negative-NPV investments (Jensen, 1986). A recent literature in dynamic capital structure argues that excessive flexibility has another cost: it induces firms to persistently increase leverage despite being suboptimal, thereby eroding the value from borrowing (Admati et al., 2018; DeMarzo and He, 2021). This problem occurs because the borrower fails to internalize how additional borrowing dilutes existing creditors' claims.

This paper studies optimal financial flexibility for firms that borrow from multiple lenders (non-exclusive lending). Financial flexibility is desirable because it allows firms to meet future liquidity needs. Meanwhile, firms face a financing constraint: they must take on initial debt to fund upfront investments. This creates incentives to subsequently take on additional debt that dilutes the initial lenders' claims, leading to over-borrowing. We show that the optimal contract imposes an endogenous borrowing limit that balances future liquidity needs against dilution incentives, even though this sometimes results in under-borrowing.

More specifically, we develop a model in which, at the initial date, a penniless borrower needs to borrow a fixed amount to finance an investment project. The project generates cash flows at the final date contingent on the borrower's effort, with higher debt reducing the borrower's incentive to exert effort. At the interim date, the borrower experiences a liquidity shock, which creates incentives

to borrow further. As a benchmark, we consider the ex post-efficient solution where liquidity shocks are observable and contractible, and the only friction is moral hazard in effort choice. We primarily focus on the case where the liquidity shock realization is the borrower's private information. The borrower can access additional borrowing after the shock; crucially, this subsequent borrowing is non-exclusive: the borrower is not obligated to return to the initial lenders. This non-exclusivity creates over-borrowing incentives and necessitates restrictions on financial flexibility.

The need for upfront borrowing requires the borrower to take on initial debt, while the liquidity shock creates incentives for additional borrowing at the interim date. Initial lenders, anticipating potential dilution from additional borrowing, naturally benefit from protections that restrict total debt. However, these restrictions can be costly when the liquidity shock is severe. We show that our problem resembles a delegation problem between a principal and an agent, where the mechanism does not allow for transfers. In this context, the principal is the borrower at the initial date, and the agent is the same borrower at the interim date after experiencing a liquidity shock. While the principal maximizes total firm value, the agent (with private information about the shock) maximizes the continuation value at the interim date, which is the firm value net of initial debt. Unlike standard delegation problems where conflict is derived from present bias, in our setting, it occurs due to the pre-existing initial debt. As Admati et al. (2018) show, this legacy debt creates a "leverage ratchet" effect where firms persistently increase leverage despite this being suboptimal.

Building on the delegation literature, we show that the optimal contract between the borrower and initial lenders includes an endogenous total debt limit. The borrower may raise additional financing at the interim date, but total debt cannot exceed a pre-specified cap. This debt limit represents what we term financial flexibility. When the liquidity shock is mild, borrowing needs are modest, and the debt limit turns out slack. The firm borrows as if under full flexibility. When the shock is severe, however, the borrower would prefer to raise more debt than the cap permits. The limit binds, which restricts the firm's access to funds. The endogenous level of the debt limit is set at the initial date to balance the value of interim liquidity provision against the need to prevent excessive borrowing and creditor dilution.

Why does the optimal contract include an endogenous debt limit? Since the liquidity shock is the borrower's private information, the optimal contract must provide incentives for truthful reporting. This requires that whenever the debt limit does not bind, the borrower can borrow exactly the amount she would choose under full flexibility. When the debt limit binds, however, the borrower cannot benefit from misreporting her liquidity shock. Under this policy, deviations in reported liquidity shock do not increase borrowing capacity because the same debt limit applies to all borrower types. This mechanism–where the borrower either receives her unconstrained optimum or faces a locally flat allocation–follows standard results from optimal delegation theory.

We show that the financing constraint always binds: the borrower raises exactly the amount needed for the initial investment and no more. This occurs because more initial borrowing comes with more debt taken at the initial date. However, due to the dilution incentives, more initial debt must be accompanied by reduced financial flexibility, i.e., a tighter debt limit, which is costly. Therefore, the borrower never raises more than necessary on the initial date. By contrast, the financing constraint might be slack under exclusive lending, where firms must return to the initial lender for additional financing. This occurs when the initial investment is relatively small. In such cases, private information, because it only arrives in the future, introduces no distortion relative to the ex post-efficient allocation.

Both over- and under-borrowing exist under the optimal contract. Borrowers who face severe liquidity shocks are constrained by the debt limit and under-borrow relative to the ex post-efficient benchmark. By contrast, borrowers experiencing mild shocks raise more debt than the ex post-efficient benchmark, resulting in over-borrowing. The endogenous debt limit is set to balance this over-borrowing and under-borrowing.

We show that the optimal contract can be implemented using a simple and commonly observed financial arrangement: an initial debt issuance paired with a covenant that limits total leverage. Such covenants are widespread in practice and, as the quotes at the beginning of the introduction illustrate, serve precisely the role highlighted in our analysis. They prevent excessive future borrowing that could dilute existing creditors. Our framework thus provides a theoretical foundation for understanding why these covenants are prevalent.

Related Literature (very preliminary)

Our paper is closely related to a strand of the mechanism design literature on commitment and flexibility (Halac and Yared, 2022, 2014; Amador et al., 2006; Amador and Bagwell, 2013). There are two main differences. First, motivated by the corporate financing settings, we incorporate both an initial financing constraint (requiring the borrower to raise a specific amount at $t = 0$) and an interim wealth constraint. We show these constraints generate important results. Second, while the time inconsistency problem in these papers comes from present bias, in our framework, it occurs because the borrower ignores the impact of her decisions on the value of old debt once it has been issued. This creates a disagreement between the principal's and the agent's objective function. The principal – the borrower before the initial debt has been issued – aims to maximize firm value, whereas the agent – the borrower after the initial debt has been issued – aims to maximize the firm value net of the initial debt. On the technical side, our solution involves a two-dimensional maximization problem: an initial debt level and a debt limit function, and the problem is not jointly convex. We manage to circumvent this by solving the problem sequentially.

Our work also connects to the finance literature on financial flexibility (Gamba and Triantis, 2008), though with different modeling primitives. These models emphasize cash hoarding for investment opportunities, whereas our paper introduces private information and focuses instead on flexibility in future leverage. The paper is also related to the theoretical models on debt covenants (Aghion and Bolton, 1992), but the methodology (mechanism design v.s. incomplete contract) differs fundamentally. Our paper is also related to the dynamic capital structure literature, which either assumes full commitment (Leland, 1994) and no commitment (DeMarzo and He, 2021) in debt issuance.

2 Model Setup

2.1 Borrower, Investment, and Liquidity Shock

There are three dates, $t = 0, 1, 2$, and a risk-neutral borrower. The borrower has a project that requires a fixed investment of I at $t = 0$ but has no resources to finance it, hence the need to borrow. The project generates no cash flow at $t = 1$ and a final cash flow Y at $t = 2$ if successful (zero otherwise).[1] At the beginning of $t = 2$, the borrower privately chooses an effort level $q \in [0, 1]$, where q corresponds to the success probability. The effort incurs a private cost $c(q)$. We assume that $c(0) = 0$ and that $c(q)$ is strictly increasing, strictly convex, and twice continuously differentiable. Moreover, we assume $c'(1) > Y$ to guarantee interior solutions for q.

The borrower does not discount cash flows between $t = 0$ and $t = 1$. At $t = 1$, she receives a liquidity shock and discounts the cash flows at $t = 2$ at θ. The realization of θ is her private information. We assume that θ is drawn from a set $\Theta = [\underline{\theta}, \bar{\theta}] \subseteq (0, 1]$, with a continuously differentiable density function $g(\theta) > 0$ and cumulative density function $G(\theta)$. Throughout the paper, we make the following standard assumption, which leads to the result that $g(\theta)/G(\theta)$ is decreasing.

Assumption 1. *The cumulative density function $G(\theta)$ is log-concave.*

2.2 Lenders and Contract

There exists a competitive pool of dispersed lenders. Throughout the paper, lenders are risk-neutral and do not discount the future. Under competition, the borrower captures all surplus from lending.

[1] Under this binary cash flow framework, there is no fundamental distinction between different securities, such as equity and debt. In addition, it implies that seniority is irrelevant within the class of debt contracts. We choose to model final cash flows in this parsimonious manner to eliminate the complicated security design problem and instead focus on financial flexibility.

The focus of this paper is to study the optimal lending contract, under which at $t = 0$, lenders make an initial transfer τ_0 to the borrower to cover the initial investment I. At $t = 1$, the borrower receives another interim transfer $\tau_1(\hat{\theta})$ at $t = 1$ contingent on the reported liquidity shock $\hat{\theta}$. If the project succeeds at $t = 2$, she will make a state-contingent payment $x_1(\hat{\theta})$. As a matter of terminology, we will interpret the repayment at $t = 2$ as a debt contract and refer to x_1 as the face value of debt. For a given shock realization θ, report $\hat{\theta}$, and effort q, the borrower's payoff is

$$\tau_0 - I + \tau_1(\hat{\theta}) + \theta \left[q \left(Y - x_1(\hat{\theta}) \right) - c(q) \right].$$

Since the borrower has no initial wealth and is protected by limited liability, the contract must satisfy the following financing and feasibility constraints:

$$\tau_0 \geq I, \qquad \tau_1(\hat{\theta}) \geq 0, \qquad x_1(\hat{\theta}) \in [0, Y]. \tag{1}$$

For the rest of the paper, we refer to the constraint $\tau_0 \geq I$ as the *financing constraint*.

Note that the borrower seeks financing on two dates: an initial investment I at $t = 0$ and an additional transfer τ_1 at $t = 1$ upon the liquidity shock. Under non-exclusive lending, *initial* lenders at $t = 0$ are separate from the *new* lenders at $t = 1$. This separation occurs naturally when lenders are dispersed because finding the same group of initial lenders at $t = 1$ can be difficult. Even if the initial lenders are concentrated (e.g., a bank), regulatory capital constraints may restrict their exposure to a particular sector or borrower, preventing them from providing additional funding at $t = 1$. Moreover, we show later that the borrower always prefers taking out debt from new lenders rather than initial lenders at $t = 1$ if given the option. Given this setup, we explicitly distinguish between debt issued to initial lenders (x_0) and that issued to new lenders ($x_1 - x_0$).

Figure 1 summarizes the sequence of events on three dates.

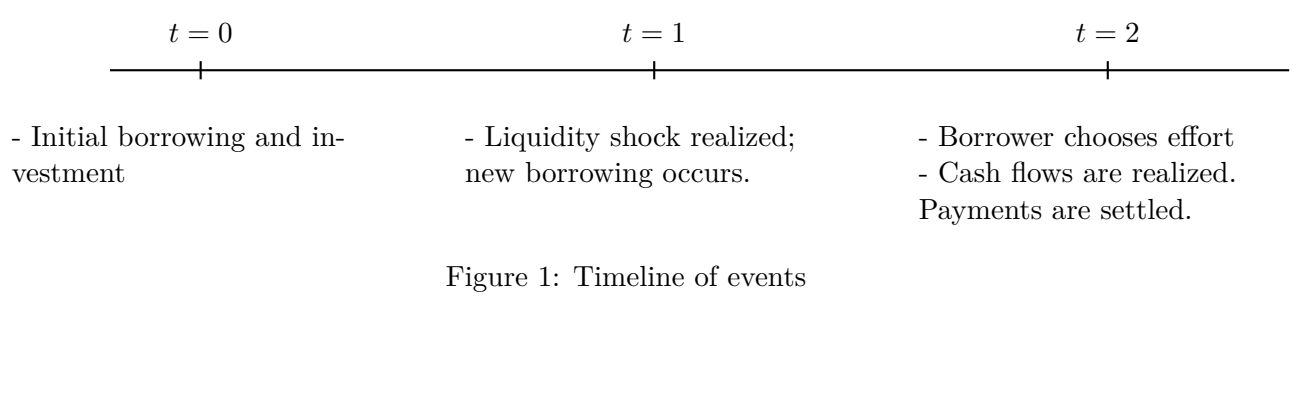

Figure 1: Timeline of events

2.3 Effort Choice and Valuations

The borrower's choice of effort at $t = 2$ involves a standard moral hazard problem, which is introduced so that leverage has a downside.[2] Given a face value of debt x_1, the borrower chooses effort q to maximize her payoff

$$V_2(x_1) = \max_{q \in [0,1]} q(Y - x_1) - c(q), \tag{2}$$

where $V_2(x_1)$ is the value of the borrower's claim net the effort cost. Because there is no outsider equity, $\theta V_2(x_1)$ can be thought of as the firm's equity value at $t = 1$. Let $q(x_1)$ be the optimal solution to this maximization problem. The following lemma summarizes some useful properties satisfied by $V_2(x_1)$ and $q(x_1)$.

Lemma 1. *Suppose that, for each x_1, $q(x_1)$ solves (2). Then*

- *$\forall x_1 \in [0, Y]$, $V_2(x_1)$ is decreasing and convex with derivative $V_2'(x_1) = -q(x_1)$.*
- *$\forall x_1 \in [0, Y]$, $q(x_1) = (c')^{-1}(Y - x_1)$.*
- *$\forall x_1 \in [0, Y]$, $q(x_1)$ is decreasing with derivative $q'(x_1) = -\frac{1}{c''(q(x_1))}$.*
- *If $c(q)$ is three times continuously differentiable, then $q''(x_1) = -\frac{c'''(q(x_1))}{c''(q(x_1))}\big(q'(x_1)\big)^2$.*

Throughout the paper, we make the following assumptions.

Assumption 2. *The cost function $c(q)$ satisfies*

$$I^{\max} \equiv \max_{x_1 \in [0,Y]} x_1 q(x_1) \geq I. \tag{3}$$

Assumption 3. *$c(q)$ is three times continuously differentiable and satisfies $c'''(q) \geq 0$.*

Assumption 2 states that the maximum expected repayment $x_1 q(x_1)$ is sufficient to cover the initial investment I. Assumption 3 and Lemma 1 imply that $q(x_1)$ is concave. This assumption will guarantee that the objective function of several of the programs we later consider is concave.

For the rest of this paper, we define

$$W_1(x_1, \theta) \equiv \underbrace{x_1 q(x_1)}_{\text{debt}} + \underbrace{\theta V_2(x_1)}_{\text{equity}}, \tag{4}$$

[2] The solution without a moral hazard problem is trivial: the borrower always pledges all the cash flows at $t = 0$. An alternative assumption is to introduce the bankruptcy cost.

as the (ex-post) $t = 1$ firm value for a given θ. Let $x_1^*(\theta)$ be the level of debt that maximizes this,

$$x_1^*(\theta) = \arg\max_{x_1} \; W_1(x_1, \theta). \tag{5}$$

This solution corresponds to the *ex post-efficient* allocation in standard mechanism design problems: it maximizes total surplus when moral hazard is the only friction. For the remainder of this paper, we also refer to this as the *efficient* debt level. This level of debt captures the trade-off between providing liquidity to the borrower and preserving her incentives to exert effort. Because $\frac{\partial^2 W_1(x_1,\theta)}{\partial\theta\partial x_1} < 0$, it is easily established that $x_1^*(\theta)$ decreases with θ: under the efficient debt level, firms with lower liquidity shocks should borrow a lower level of total debt.

Note that the ex post-efficient allocation corresponds to the solution after knowing θ. If the borrower chooses at $t = 0$ without knowing θ, the optimal choice is $x_1^*(\mathbb{E}[\theta])$, because $W_1(x_1, \theta)$ is linear in θ. For the remainder of this paper, we refer to $x_1^*(\mathbb{E}[\theta])$ as the ex-ante optimal debt.

Linear Example. *Throughout our analysis, we study a specific example where the cost function $c(q)$ is quadratic. Under this specification, the optimal effort $q(x_1)$ is linear in x_1, and the borrower's payoff at $t = 2$ is quadratic. For the rest of this paper, we will refer to this case as the linear example. Specifically, if $c(q) = \frac{1}{2}cq^2$ where $c > Y$, then*

$$\begin{aligned} q(x_1) &= \frac{Y - x_1}{c} \\ V_2(x_1) &= \frac{1}{2}\frac{(Y - x_1)^2}{c}. \end{aligned} \tag{6}$$

Assumption 2 is satisfied only if

$$\frac{Y^2}{4c} \geq I.$$

Moreover, we have

$$\begin{aligned} W_1(x_1, \theta) &= x_1\frac{Y - x_1}{c} + \frac{\theta}{2}\frac{(Y - x_1)^2}{c} \\ x_1^*(\theta) &= \frac{1 - \theta}{2 - \theta}Y. \end{aligned}$$

2.4 Complete Information Solution

If θ is observable and contractible, the optimal long-term contract is designed to maximize the borrower's expected payoff, subject to the lender's participation constraint, as well as financing and

feasibility constraints.

$$
\begin{aligned}
&\max_{\{\tau_0,\tau_1(\theta),x_1(\theta)\}} \quad \tau_0 - I + \mathbb{E}\Big[\tau_1(\theta) + \theta V_2\left(x_1(\theta)\right)\Big] \\
&\text{subject to} \\
&\mathbb{E}\Big[x_1(\theta)q\left(x_1(\theta)\right) - \tau_1(\theta)\Big] - \tau_0 \geq 0 \\
&\tau_0 \geq I, \qquad \tau_1(\theta) \geq 0, \qquad x_1(\theta) \in [0, Y].
\end{aligned} \tag{7}
$$

Clearly, the lender's participation constraint is always binding, and it is without loss of generality to set $\tau_1(\theta) = 0$. We substitute the participation constraint in the objective function and formulate the Lagrangian. Let λ be the Lagrange multiplier of the financing constraint. Dividing the Lagrangian by $1 + \lambda$, the problem becomes

$$
\min_{\lambda \geq 0} \max_{\{x_1(\theta)\}} \ \mathbb{E}\Big[W_1\left(x_1(\theta), \theta\right) - \frac{\lambda}{1+\lambda}\theta V_2\left(x_1(\theta)\right)\Big] - I, \tag{8}
$$

Proposition 1 (Complete Information Benchmark)**.** *Let*

$$
I^c \equiv \mathbb{E}\Big[x_1^*(\theta)q\left(x_1^*(\theta)\right)\Big].
$$

When θ is observable and contractible, it is without loss of generality to set $\tau_1(\theta) = 0$.

- *If $I^c \geq I$, the financing constraint is slack, and the optimal solution satisfies $x_1^{c*}(\theta) = x_1^*(\theta)$ and $\tau_0 = I^c$.*
- *Otherwise, the solution is $x_1^{c*}(\theta) = x_1^*\left(\delta^{c*}\theta\right)$ where $\delta^{c*} \equiv 1/(1+\lambda^{c*})$ solves*

  $$
  \mathbb{E}\left[x_1^*\left(\delta^{c*}\theta\right) q\left(x_1^*\left(\delta^{c*}\theta\right)\right)\right] = I.
  $$

  In this case, $\tau_0 = I$.

The term $\mathbb{E}[x_1^*(\theta)q(x_1^*(\theta))]$ measures the ex-ante pledgeable income at $t = 0$ under the efficient debt level. Lemma 1 presents two distinct cases, depending on whether this pledgeable income is sufficient to cover the initial investment I. When sufficient, the financing constraint is slack, and the efficient debt level is optimal. When insufficient, the financing constraint binds, so the borrower must pledge more. The second case of Lemma 1 shows this scenario is equivalent to the efficient debt level where the borrower effectively faces a more severe liquidity shock $\delta^{c*}\theta$. In other words, the financing constraint amplifies the borrower's liquidity shock when it binds. As a result, this leads to a level of total debt higher than the ex post efficient allocation.

We note that the results are equivalent whether lending is exclusive or not. Intuitively, when total debt can be contracted without incurring incentive rents, future borrowing can be perfectly constrained and is used only when needed. Therefore, it does not matter who holds the debt.

We conclude this subsection by making a note on lenders' ex-post returns: while lenders break even in expectation, their ex-post profits depend on the realized value of θ. Specifically, lenders incur losses when θ is high but earn profits when θ is low. The intuition is straightforward: borrowers with low θ value less of the cash flows at $t = 2$, so that it is relatively cheaper for them to increase their debt burden (i.e., to choose higher x_1) compared to borrowers with high θ. As a result, lenders profit from these borrowers. This finding contrasts with Holmström and Tirole (1998), where lenders lose money when the realized liquidity shock—modeled as a liquidity need—is high.

Linear Example. *In the case of complete information, it is easily derived that*

$$\mathbb{E}\left[x_1^*\left(\delta\theta\right) q\left(x_1^*\left(\delta\theta\right)\right)\right] = \frac{Y^2}{c}\mathbb{E}\left[\frac{1-\delta\theta}{\left(2-\delta\theta\right)^2}\right].$$

If we further consider the special case of the uniform distribution: $G(\theta) = \frac{\theta-\underline{\theta}}{\bar{\theta}-\underline{\theta}}$, *then*

$$\frac{Y^2}{c}\mathbb{E}\left[\frac{1-\delta\theta}{\left(2-\delta\theta\right)^2}\right] = \frac{1}{\bar{\theta}-\underline{\theta}}\frac{Y^2}{\delta c}\left[\frac{1}{2-\delta\underline{\theta}} - \frac{1}{2-\delta\bar{\theta}} - \log\left(\frac{2-\delta\bar{\theta}}{2-\delta\underline{\theta}}\right)\right].$$

The financing constraint is slack if

$$\frac{1}{\bar{\theta}-\underline{\theta}}\frac{Y^2}{c}\left[\frac{1}{2-\underline{\theta}} - \frac{1}{2-\bar{\theta}} - \log\left(\frac{2-\bar{\theta}}{2-\underline{\theta}}\right)\right] \geq I.$$

3 Benchmark: Single Lender Equilibrium

Before solving the model, we first analyze a benchmark case where the borrower can commit to a long-term contract with a single lender, whose payoff is

$$x_1(\theta)q\left(x_1(\theta)\right) - \tau_1(\theta) - \tau_0.$$

This corresponds to the case of exclusive lending, and we refer to the solution as the information-constrained optimal allocation. In addition to moral hazard, the remaining frictions are the financing constraint and the borrower's private information about her liquidity shock.

By the revelation principle, we can restrict attention to direct revelation mechanisms $\{\tau_0, \tau_1(\theta), x_1(\theta)\}$. The optimal mechanism can be found by solving the problem in (7) with the additional truth-telling

IC constraint

$$\theta \in \arg\max_{\hat{\theta}} \ \tau_1(\hat{\theta}) + \theta V_2(x_1(\hat{\theta})). \tag{9}$$

Using the envelope theorem, we can rewrite (9) as

$$\begin{aligned} &\tau_1(\theta) + \theta V_2\left(x_1(\theta)\right) = \tau_1(\bar{\theta}) + \bar{\theta} V_2(x_1(\bar{\theta})) - \int_{\theta}^{\bar{\theta}} V_2(x_1(\tilde{\theta}))d\tilde{\theta}, \ \forall \theta \in \Theta \\ &x_1(\theta) \text{ is non-increasing.} \end{aligned} \tag{10}$$

As in the problem under complete information, the lender's $t = 0$ participation constraint must be binding. Simple calculation shows that the initial financing constraint becomes [3]

$$\tau_0 = \mathbb{E}\left[W_1(x_1(\theta), \theta) + \frac{G(\theta)}{g(\theta)} V_2\left(x_1(\theta)\right)\right] - \tau_1(\bar{\theta}) - \bar{\theta} V_2(x_1(\bar{\theta})) \geq I. \tag{11}$$

Substituting τ_0 and $\tau_1(\theta)$ into the objective function, the optimal contract solves[4]

$$\begin{aligned} &\max_{\{x_1(\theta), \tau_1(\theta) \geq 0\}} \ \mathbb{E}\left[W_1(x_1(\theta), \theta)\right] - I \\ &\text{subject to (10) and (11)} \end{aligned} \tag{12}$$

We make the following assumption to ensure that the previous optimization problem is convex:

Assumption 4. *For all $\theta \in \Theta$, the function*

$$W_1(x_1, \theta) + \frac{G(\theta)}{g(\theta)} V_2(x_1)$$

is concave in x_1. Under Assumption 1, this holds if

$$2 + \frac{x_1 q''(x_1)}{q'(x_1)} \geq \bar{\theta} + \frac{1}{g(\bar{\theta})}.$$

Note that problem (12) differs from the standard mechanism design problem in that it has an additional constraint $\tau_1(\theta) > 0$. Lemma 3 in the appendix proves that $\tau_1(\theta)$ must be non-

[3] This involves the following simplification by changing the order of integration:

$$\int_{\underline{\theta}}^{\bar{\theta}} \int_{\theta}^{\bar{\theta}} V_2(x_1(\tilde{\theta}))g(\theta)d\theta = \int_{\underline{\theta}}^{\bar{\theta}} \int_{\underline{\theta}}^{\tilde{\theta}} V_2(x_1(\tilde{\theta}))g(\theta)d\theta d\tilde{\theta} = \int_{\underline{\theta}}^{\bar{\theta}} V_2(x_1(\tilde{\theta}))G(\tilde{\theta})d\tilde{\theta}.$$

The problem also substitutes the envelope condition (10) in the participation constraint.

[4] The constraint $x_1(\theta) \in [0, Y]$ will always be slack and hence omitted.

increasing. Intuitively, borrowers with larger liquidity shocks have incentives to borrow more. This result implies that we only need to evaluate the constraint at $\theta = \bar{\theta}$. In addition, it is optimal to set $\tau_1(\bar{\theta}) = 0$. Otherwise, the borrower can reduce $\tau_1(\theta)$ for all θ by the same amount and borrow more at $t = 0$, which relaxes the financing constraint.[5]

We solve the problem in two steps. First, with some slight abuse of notation, let λ be the Lagrange multiplier of the financing constraint $\tau_0 \geq I$ and $\delta \equiv 1/(1+\lambda)$ be the shadow discount factor. We find the optimal solution for a fixed value of δ:

$$\begin{aligned} &\max_{\{x_1(\theta)\}} \ \mathbb{E}\left[W_1(x_1(\theta),\theta) + (1-\delta)\frac{G(\theta)}{g(\theta)}V_2(x_1(\theta))\right] - (1-\delta)\bar{\theta}V_2(x_1(\bar{\theta})) \\ &\text{subject to} \\ &x_1(\theta) \text{ is non-increasing.} \end{aligned} \tag{13}$$

Define

$$x_1^s(\theta) \equiv \arg\max_{x_1} \left\{W_1(x_1,\theta) + (1-\delta)\frac{G(\theta)}{g(\theta)}V_2(x_1)\right\}, \tag{14}$$

which maximizes point-wise the virtual surplus. Note that $x_1^s(\theta) = x_1^*(\theta)$ when $\delta = 1$. Under Assumption 1, the objective function in (14) satisfies the single-crossing property, so that $x_1^s(\theta)$ is non-increasing in θ. We cannot apply the standard pointwise maximization technique because the objective function in equation (13) includes a final term involving $\bar{\theta}$.[6] Intuitively, even for the most patient borrower type $\bar{\theta}$, the mechanism cannot fully extract rents due to the limited liability constraint $\tau_1(\theta) \geq 0$. The IC constraint (10) would then imply that a similar term accrues to all borrower types.

By applying the method from the global theory of constrained optimization, we have the following result.

Proposition 2. *Let $x_1^{s*}(\theta)$ be the solution to problem (13) for a given δ.*

- *If $\delta \geq \underline{\theta}/\mathbb{E}[\theta]$, the solution to problem (13) is $x_1^{s*}(\theta) = \max\{x_1^s(\theta), x_1^s(\theta_H)\}$ where $\theta_H \in \Theta$*

[5] If the financing constraint is slack, the borrower is indifferent between borrowing at $t = 0$ and $t = 1$ due to no discount between the two dates. As long as there is a slight preference for $t = 0$ borrowing, $\tau_1(\bar{\theta}) = 0$ would hold.

[6] To see this, note that $x_1(\bar{\theta})$ is chosen to maximize

$$W_1(x_1(\bar{\theta}),\bar{\theta}) + (1-\delta)\frac{G(\bar{\theta})}{g(\bar{\theta})}V_2(x_1(\bar{\theta})) - (1-\delta)\bar{\theta}V_2(x_1(\bar{\theta})),$$

so that the point-wise maximum solution will lead to a solution where $x_1(\theta)$ has an upward jump at $\bar{\theta}$ unless $\delta = 1$.

solves:

$$\int_{\theta_H}^{\bar{\theta}} \frac{\partial W_1(x_1^s(\theta_H),\theta)}{\partial x_1} dG(\theta) + (1-\delta) q\left(x_1^s(\theta_H)\right) \left(\theta_H G(\theta_H) + \int_{\theta_H}^{\bar{\theta}} \theta dG(\theta)\right) = 0. \tag{15}$$

- *If* $\delta < \underline{\theta}/\mathbb{E}[\theta]$, $x_1^{s*}(\theta) = x_1^p$, *where* x_1^p *solves*

$$x_1^p q'(x_1^p) + (1 - \delta\mathbb{E}[\theta])\, q(x_1^p) = 0.$$

The optimal transfers $\{\tau_0, \tau_1(\theta)\}$ *satisfy* (10) *and* (11).

Proposition 2 describes the optimal level of debt under different degrees of financing constraint. When $\delta \geq \underline{\theta}/\mathbb{E}[\theta]$, the financing constraint is not too tight and may even be slack. In this case, the optimal debt level decreases with θ for lower values of θ (following $x_1^s(\theta)$) and then remains constant at $x_1^s(\theta_H)$ once θ exceeds the threshold θ_H. In contrast, when $\delta < \underline{\theta}/\mathbb{E}[\theta]$, the financing constraint becomes very tight. In this case, the optimal debt level is a constant at x_1^p across all types θ.

Borrowers hit by lower liquidity shocks (higher θ) inherently need to borrow less at $t = 1$. The incentive compatibility constraint then requires that the interim payment τ_1 must also decrease for these borrowers. This creates a natural downward-sloping pattern in both the total debt level and interim transfers as θ increases. Given the financing constraint, the principal – the borrower at $t = 0$ – prefers to pledge more of the future cash flows, which necessarily involves a lower τ_1. Because τ_1 decreases with θ, this eventually causes borrowers with smaller liquidity shocks to hit the limited liability constraint $\tau_1(\theta) \geq 0$. Once this constraint binds for types above θ_H, these borrowers cannot receive any interim payment ($\tau_1 = 0$). To maintain incentive compatibility when the limited liability constraint binds, the face value of debt x_1 must also remain constant for all types above θ_H.

We now proceed to the second step.

Proposition 3. *Let*

$$I^s \equiv \mathbb{E}\left[W_1\left(x_1^*(\theta),\theta\right) + V_2\left(x_1^*(\theta)\right)\frac{G(\theta)}{g(\theta)}\right] - \bar{\theta} V_2\left(x_1^*(\bar{\theta})\right),$$

where $x_1^*(\theta)$ *is defined in* (5). *Then,*

- *If* $I \leq I^s$, *the optimal policy is* $x_1^*(\theta)$, *and* $\delta^{s*} = 1$.
- *If* $I > I^s$, *the optimal policy is* $x_1^{s*}(\theta)$, *where* $\{\theta_H, \delta^{s*}\}$ *jointly solve equations* (15) *and* $\tau_0 = I$.

It is easily shown that $I^s = I^c - \mathbb{E}[\tau_1(\theta)]$. Intuitively, when lenders expect to make a transfer $\tau_1(\theta)$ at $t = 1$ due to incomplete information, they will reduce the initial lending by the expected amount $\mathbb{E}[\tau_1(\theta)]$ compared to the complete-information benchmark. Proposition 3 characterizes how the optimal solution depends on the investment size and, hence, the financing needs. When the financing constraint is slack (which corresponds to $I \leq I^s$), the optimal solution is identical to the efficient debt level, i.e., the borrower's private information about θ does not introduce any additional distortion in the debt choice. This is because private information has not yet been realized at $t = 1$, the time of contracting. As a result, there is no need to provide information rents, and the allocation involves no distortion (Baron and Besanko, 1984).

However, when the financing constraint binds (i.e., when $I > I^s$), the truth-telling incentive compatibility constraint introduces distortions. As shown by (14), the extent of this distortion depends on δ, which captures the severity of the financing constraint. The solution then is directly linked to the two cases in Proposition 2, where the face value of debt involves a flat region.

Linear Example. *In the single-lender equilibrium, we have*

$$x_1^s(\theta) = Y \left[\frac{1 - \theta - (1-\delta)\frac{G(\theta)}{g(\theta)}}{2 - \theta - (1-\delta)\frac{G(\theta)}{g(\theta)}} \right].$$

If $G(\theta)$ is log-concave, we have that Assumption 4 holds if

$$(2 - \bar{\theta})g(\bar{\theta}) \geq 1.$$

Substituting in the envelope condition, we get that for $\theta \leq \theta_H$ the transfer to the borrower at $t = 1$ is

$$\tau_1(\theta) = \frac{Y^2}{2c} \left[\frac{\theta_H}{\left(2 - \theta_H - (1-\delta)\frac{G(\theta_H)}{g(\theta_H)}\right)^2} - \frac{\theta}{\left(2 - \theta - (1-\delta)\frac{G(\theta)}{g(\theta)}\right)^2} - \int_\theta^{\theta_H} \frac{1}{\left(2 - \tilde{\theta} - (1-\delta)\frac{G(\tilde{\theta})}{g(\tilde{\theta})}\right)^2} d\tilde{\theta} \right].$$

Moreover, we get

$$x_1^P = \frac{1 - \delta\mathbb{E}[\theta]}{2 - \delta\mathbb{E}[\theta]}.$$

If we further assume that $\theta \sim U[\underline{\theta}, \bar{\theta}]$, we get that

$$x_1^s(\theta) = Y\left[1 - \frac{1}{\alpha - \beta\theta}\right]$$
$$\tau_1(\theta) = \frac{Y^2}{2c}\left[\frac{\theta_H}{(\alpha - \beta\theta_H)^2} - \frac{\theta}{(\alpha - \beta\theta)^2} - \frac{\theta_H - \theta}{(\alpha - \beta\theta_H)(\alpha - \beta\theta)}\right]$$

where $\alpha \equiv 2 + (1-\delta)\underline{\theta}$ and $\beta = 2 - \delta$.

4 Equilibrium under Non-Exclusive Lending

We now solve the model under non-exclusive lending. In this setting, initial lenders anticipate dilution from new debt issued at $t = 1$ and, therefore, seek to limit future issuance to protect their claims. We consider the contract $\{\tau_0, x_0, x_1(\hat{\theta})\}$ offered to initial lenders, where total debt is capped by $x_1(\hat{\theta})$, a function of the borrower's reported type $\hat{\theta}$ at $t = 1$. By doing so, we restrict the mechanism to those where transfers cannot be made to the initial lenders and adjustments cannot be applied to x_0 conditional on the reported type $\hat{\theta}$ at $t = 1$. We examine such mechanisms later in Section 5.

Since the borrower faces a liquidity shock, it is without loss of generality to assume she issues new debt up to the debt limit $x_1(\hat{\theta})$. Thus, $\Delta x_1(\hat{\theta}) \equiv x_1(\hat{\theta}) - x_0$ is the additional debt issued at $t = 1$.

4.1 Reduction to Delegation Problem

We first show that the problem can be equivalently formulated as a delegation problem (Amador et al., 2006; Amador and Bagwell, 2013). Because the borrower offers the contract, new lenders' participation constraint must bind, implying $\tau_1 = (x_1 - x_0)q(x_1)$, where $x_1 \geq x_0$ is required to satisfy the non-negativity constraint $\tau_1 \geq 0$. At $t = 1$, the borrower's continuation payoff for any given (θ, x_1) is:

$$V_1(x_1, x_0, \theta) = \underbrace{(x_1 - x_0)q(x_1)}_{\text{new debt issuance}} + \theta V_2(x_1) = W_1(x_1, \theta) - x_0 q(x_1). \tag{16}$$

Meanwhile, the participation constraint for initial lenders must also bind, implying

$$\tau_0 = x_0 \mathbb{E}\left[q(x_1(\theta))\right],$$

and the financing constraint becomes

$$x_0 \mathbb{E}\left[q(x_1(\theta))\right] \geq I. \tag{17}$$

Therefore, the borrower's expected payoff at $t = 0$ is given by:

$$\tau_0 - I + \mathbb{E}\left[V_1(x_1(\theta), x_0, \theta)\right] = \mathbb{E}\left[W_1(x_1, \theta)\right] - I,$$

so that the objective is to maximize $\mathbb{E}\left[W_1(x_1, \theta)\right]$. The borrower has incentives to report its type truthfully only if the following incentive compatibility constraint is satisfied

$$\theta \in \arg\max_{\hat{\theta} \in \Theta} \; V_1(x_1(\hat{\theta}), x_0, \theta). \tag{18}$$

By the envelope theorem, this can be rewritten as

$$\begin{aligned} &V_1(x_1(\theta), x_0, \theta) = V_1(x_1(\bar{\theta}), x_0, \bar{\theta}) - \int_{\theta}^{\bar{\theta}} V_2(x_1(\tilde{\theta}))d\tilde{\theta} \quad \forall \theta \in \Theta \\ &x_1(\theta) \text{ is non-increasing.} \end{aligned} \tag{19}$$

The optimal mechanism at $t = 0$ solves the following problem:

$$\begin{aligned} &\max_{x_0, x_1(\theta) \geq x_0} \mathbb{E}\Big[W_1(x_1, \theta)\Big] \\ &\text{subject to (17) and (19).} \end{aligned} \tag{20}$$

This problem relates to a delegation setup, where the principal is the borrower at time $t = 0$, and the agent is the borrower at time $t = 1$. The principal delegates the choice of total debt to the agent, who has private information about the liquidity shock θ. The principal seeks to maximize firm value (net investment I), while the agent maximizes $V_1(x_1, x_0, \theta)$, which, as shown by equation (16), is the firm value net of initial debt at $t = 1$. This conflict relates to the insight of Aguiar et al. (2019), which shows that equilibrium debt issuance decisions can be characterized as the solution to a planner's problem that ignores the payoff to existing creditors.

Our problem differs in several aspects. First, it includes a financing constraint (17), which arises naturally in corporate finance settings. This constraint is later shown to always bind, adding both new insights and analytical complexity to the problem. Second, unlike Amador and Bagwell (2013), where the time-inconsistency problem is driven by present bias, here it exists because the agent disregards the debt issued at $t = 0$, a concern raised by Fama and Miller (1972) and Black

and Scholes (1973) and formalized by Admati et al. (2018). Specifically, note that

$$W_1(x_1, \theta) - V_1(x_1, x_0, \theta) = x_0 q(x_1),$$

so the wedge between the principal's and agent's payoffs exactly equals the value of the initial debt.

The optimization problem in (20) is not jointly convex in x_0 and $x_1(\theta)$. Hence, we analyze the problem in two steps. In subsection 4.2, we characterize the optimal debt at $t = 1$ for an exogenously given x_0. In subsection 4.3, we solve for the amount of initial debt x_0 issued at $t = 0$.

4.2 Constrained Debt Limits

Throughout this subsection, we take the initial debt level x_0 as given. Let us define the value function

$$\begin{aligned} W_0(x_0) = \max_{x_1(\theta) \geq x_0} \mathbb{E}\Big[W_1(x_1, \theta)\Big] \\ \text{subject to (17) and (19).} \end{aligned} \tag{21}$$

Before proceeding, it is helpful to consider a local version of the incentive compatibility constraint, which applies at any point θ where $x_1(\theta)$ is differentiable. Specifically, the first-order condition of (18) implies

$$\frac{\partial V_1(x_1(\theta), x_0, \theta)}{\partial x_1} \cdot \frac{dx_1(\theta)}{d\theta} = 0. \tag{22}$$

This condition implies that a necessary condition for truthful reporting is either: (i) the borrower receives the unconstrained optimal debt level $x_1(\theta)$ that maximizes her continuation payoff $V_1(x_1(\hat{\theta}), x_0, \theta)$, or (ii) the allowed debt level $x_1(\theta)$ is locally insensitive to the type reported, so the agent does not benefit from locally misreporting the liquidity shock. Regarding the first case, let us define

$$x_1^f(\theta, x_0) = \arg\max_{x_1} \; V_1(x_1, x_0, \theta) \tag{23}$$

as the optimal total debt under *flexible* debt issuance. For notational simplicity, we omit the dependence on x_0 and write $x_1^f(\theta)$ when no confusion arises. The first-order condition is:[7]

$$(x_1^f(\theta) - x_0) q'(x_1^f(\theta)) + (1 - \theta) q(x_1^f(\theta)) = 0. \tag{24}$$

The next lemma summarizes properties of the flexible-debt solution used throughout the analysis.

Lemma 2. *The flexible-debt solution $x_1^f(\theta, x_0)$ satisfies the following properties:*

[7] The function $V_1(x_1, x_0, \theta)$ is concave in x_1 under Assumption 3.

1. *$x_1^f(\theta, x_0)$ is nonincreasing in θ and nondecreasing in x_0.*

2. *$x_1^f(\theta, x_0) \geq x_1^*(\theta)$.*

3. *$x_1^f(\theta, x_0) \geq x_0$. If, $\bar{\theta} = 1$, then $x_1^f(\bar{\theta}, x_0) = x_0$.*

To simplify the exposition, we make the following assumption:

Assumption 5. $\bar{\theta} = 1$.

Under Assumption 5, Lemma 2 establishes that the most patient borrower does not issues any debt at $t = 1$, so $x_1^f(\bar{\theta}) = x_0$. This assumption eliminates the need to consider limits that constrain all types, reducing the cases to analyze.[8] Moreover, since Lemma 2 establishes that $x_1^f(\theta, x_0)$ is nonincreasing in θ, for any debt limit between x_0 and $x_1^f(\underline{\theta})$, there exists a threshold type $\theta_L \in \Theta$ such that $x_1^f(\theta_L)$ is exactly this limit. In equilibrium, some borrowers hit the debt limit, allowing us to specify the limit equivalently as a threshold type θ_L.

We now solve (21) and establish conditions under which the optimal mechanism takes the form of a debt limit $x_1^f(\theta_L)$ so that the borrower has total debt $x_1^L(\theta) \equiv \min\{x_1^f(\theta), x_1^f(\theta_L)\}$. By construction, this debt policy satisfies the local incentive compatibility conditions (22) and is also globally incentive compatible (i.e., satisfies (19)). Figure 2 plots the debt policy (solid red line) against the flexible debt level $x_1^f(\theta)$ (dashed red line) and the efficient debt level $x_1^*(\theta)$ (solid green line). Several patterns emerge. First, by Lemma 2, $x_1^f(\theta)$ is nonincreasing in θ, so the debt limit binds at low θ but becomes slack as θ rises. Second, Lemma 2 also gives $x_1^f(\theta) \geq x_1^*(\theta)$, since the former maximizes the firm value net of initial debt, while the latter maximizes total firm value. As a result, the debt limit leads to under-borrowing at low θ and over-borrowing at high θ.

The optimal debt limit can thus be found by solving:

$$\begin{aligned} &\max_{\theta_L \in \Theta} \ \mathbb{E}\left[W_1\left(x_1^f(\theta_L), \theta\right) \mathbf{1}_{\theta \leq \theta_L} + W_1\left(x_1^f(\theta), \theta\right) \mathbf{1}_{\theta > \theta_L}\right] \\ &\text{subject to} \\ &x_0 \mathbb{E}\left[q\left(\min\{x_1^f(\theta), x_1^f(\theta_L)\}\right)\right] \geq I. \end{aligned} \tag{25}$$

Again, we formulate Lagrangian. Let λ be the multiplier and define $\delta = \frac{1}{\lambda+1}$. Since q is decreasing and Lemma 2 establishes that $x_1^f(\theta, x_0)$ is nonincreasing in θ, it is easily established that $\mathbb{E}\left[q\left(\min\{x_1^f(\theta), x_1^f(\theta_L)\}\right)\right]$ increases in θ_L. This means that we only need to consider two candidates for the solution of problem (25): either (i) the constraint is slack, and θ_L is chosen to

[8] When $\bar{\theta} < 1$, we would need to consider additional cases to allow for debt limits of the form $0 < x_1^L < x_1^f(\bar{\theta})$. This corresponds to the degenerated delegation sets in Amador et al. (2018). The analysis in the appendix considers this more general case and allows for $\bar{\theta} < 1$.

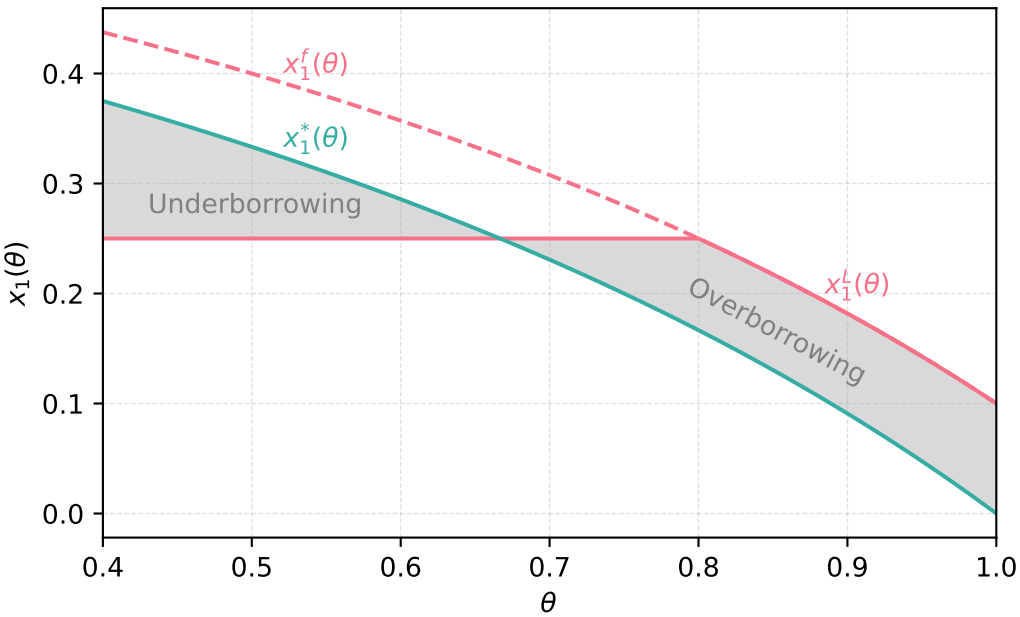

Figure 2: Debt Limit and Efficient Debt Level. We assume that $\theta \sim U[0.4, 1]$, $Y = 1$ and $c(q) = q^2$. For the purpose of illustration, we take $x_0 = 0.1$ and $\theta_L = 0.8$.

maximize the objective function; or (ii) the constraint is binding and θ_L is determined by the financing constraint. The following proposition characterizes the solution to problem (25).

Proposition 4 (Optimal Debt Limit)**.** *Let θ_L^{bc} be the lowest type that satisfies the financing constraint:*

$$\theta_L^{\text{bc}} \equiv \min\left\{\theta_L \in \Theta : \, x_0 \mathbb{E}\left[q\big(x_1^f(\theta_L)\big)\, \mathbf{1}_{\{\theta \le \theta_L\}} + q\big(x_1^f(\theta)\big)\, \mathbf{1}_{\{\theta > \theta_L\}}\right] \ge I\right\}.$$

Let θ_L^{uc} be the optimal threshold in the absence of the financing constraint.

- *If $x_0 > x_1^*\big(\mathbb{E}[\theta]\big)$, then $\theta_L^{\text{uc}} = \bar{\theta} = 1$.*
- *If $x_0 \le x_1^*\big(\mathbb{E}[\theta]\big)$, then θ_L^{uc} solves*

$$x_0 \cdot \frac{q'\big(x_1^f(\theta_L)\big)}{q\big(x_1^f(\theta_L)\big)} + \mathbb{E}[\theta_L - \theta \mid \theta \le \theta_L] = 0. \tag{26}$$

The optimal threshold is $\theta_L = \max\left\{\theta_L^{\text{uc}}, \theta_L^{\text{bc}}\right\}$, and the corresponding debt limit is $x_1^L = x_1^f(\theta_L)$.

- *If $\theta_L^{\text{uc}} > \theta_L^{\text{bc}}$, the financing constraint is slack, and $\delta = 1$.*

- *If* $\theta_L^{\text{uc}} \leq \theta_L^{\text{bc}}$*, the financing constraint binds, and* $\delta = -\frac{x_0}{\mathbb{E}[\theta_L^{\text{bc}} - \theta | \theta \leq \theta_L^{\text{bc}}]} \cdot \frac{q'\left(x_1^f(\theta_L^{\text{bc}})\right)}{q\left(x_1^f(\theta_L^{\text{bc}})\right)}$.

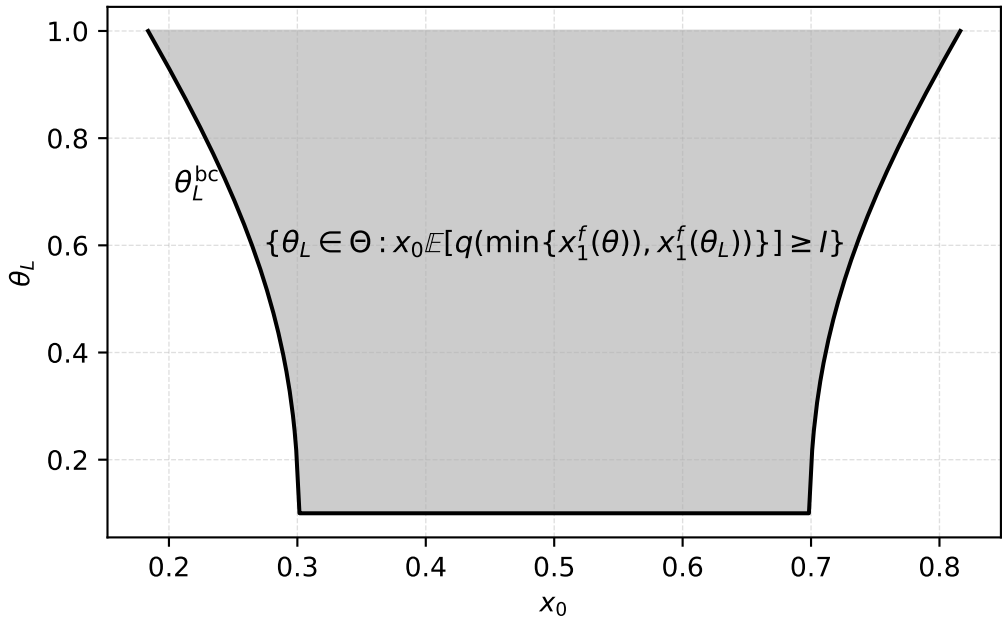

Figure 3: Illustration feasible set for the debt limit problem (25). We assume that $\theta \sim U[0.1, 1]$, $Y = 1$ and $c(q) = q^2$. The maximum pledgeable income is $I = 0.6 \times I^{\max}$ where $I^{\max} = \max_{x_1 \in [0,Y]} \; x_1 q(x_1)$.

Figure 3 illustrates the set of (x_0, θ_L) where the borrower can raise at least I at $t = 0$. The shaded region is the feasible set, with lower boundary being θ_L^{bc}. A minimum x_0 exists below which raising I is impossible. When x_0 is low but above this minimum, increasing x_0 decreases θ_L^{bc}, effectively raising the debt limit at $t = 1$. This happens because to raise I, a higher x_0 can be accommodated with a lower success probability, which corresponds to a higher level of total debt. As x_0 increases further, the constraint becomes slack, because the maximum feasible debt exceeds what even type $\underline{\theta}$ would flexibly take. At very high x_0, the pattern reverses: further increases raise θ_L^{bc}, effectively tightening the limit. This occurs because high x_0 already creates severe moral hazard in effort choice, reducing initial debt's market value. To preserve this value, a tighter debt is limit is needed so that the expected payment to initial lenders can be at least I. In later analysis, the region where θ_L^{bc} increases with x_0 is never optimal in the borrower's problem.

Meanwhile, the threshold θ_L^{uc} balances over- and under-borrowing when the financing constraint is ignored. Without knowing θ, the optimal debt level is $x_1^*\big(\mathbb{E}[\theta]\big)$ at $t = 0$. If initial debt x_0 exceeds $x_1^*(\mathbb{E}[\theta])$, there is already over-borrowing at $t = 0$, so θ_L^{uc} is chosen to impose the tightest possible

constraint to reduce over-borrowing. When x_0 falls below $x_1^*(\mathbb{E}[\theta])$, θ_L^{uc} is determined by equation (26), maximizing the objective in (25). As Figure 2 shows, such debt limit causes under-borrowing when θ is low but over-borrowing when θ is high. The threshold θ_L minimizes the combined welfare loss from both distortions.

According to Proposition 4, the optimal policy takes $\theta_L = \max\{\theta_L^{\text{uc}}, \theta_L^{\text{bc}}\}$, acknowledging that the threshold must be at least θ_L^{bc} to satisfy the initial financing requirement. When $\theta_L^{\text{uc}} > \theta_L^{\text{bc}}$, the financing constraint does not bind ($\lambda = 0$), whereas when $\theta_L^{\text{uc}} \leq \theta_L^{\text{bc}}$, the constraint binds and generates a positive Lagrange multiplier reflecting the shadow cost of the financing requirement.

The next proposition identifies sufficient conditions under which the debt limit characterized in the proposition solves the mechanism design problem (33). Our proof follows the Lagrangian methodology employed by Amador et al. (2006) and Amador and Bagwell (2013).

Proposition 5 (Sufficient Conditions for Optimality)**.** *Consider the debt limit in Proposition 4. Let $\delta = 1/(1+\lambda)$ where λ be the Lagrange multiplier of the financing constraint in (25). If the function*

$$\delta G(\theta) + \frac{x_0 q'(x_1^f(\theta))}{q(x_1^f(\theta))} g(\theta) \tag{27}$$

is non-decreasing for $\theta \in [\theta_L, \bar{\theta}]$. Then, $x_1^{n*}(\theta) = \min\{x_1^f(\theta), x_1^f(\theta_L)\}$ solves the mechanism design problem (21).

Why is a debt limit optimal? The agent ignores how additional borrowing dilutes existing debt x_0, leading to over-borrowing. The principal naturally wants to impose discipline. Since the liquidity shock is private information, the agent's incentive compatibility constraint requires that total debt either maximizes the agent's continuation value or is locally flat in the reported type (equation (22)). Condition (27) is a regularity condition that ensures this is also globally optimal.

Linear Example. *Under non-exclusive lending, we have*

$$V_1(x_1, x_0, \theta) = (x_1 - x_0)\frac{Y - x_1}{c} + \frac{\theta}{2}\frac{(Y - x_1)^2}{c}$$
$$x_1^f(\theta, x_0) = \frac{(1-\theta)Y + x_0}{2-\theta}.$$

The price at $t = 1$ given $x_1^f(\theta, x_0)$ is

$$q(x_1^f(\theta, x_0)) = \frac{Y - x_0}{c}\frac{1}{2-\theta},$$

so

$$x_0\mathbb{E}\left[q\left(\min\{x_1^f(\theta), x_1^f(\theta_L)\}\right)\right] = \frac{x_0(Y-x_0)}{c}\mathbb{E}\left[\frac{1}{2-\theta_L}\mathbf{1}_{\{\theta\leq\theta_L\}} + \frac{1}{2-\theta}\mathbf{1}_{\{\theta>\theta_L\}}\right].$$

The sufficient condition (27) *in Proposition* 5 *is satisfied only if*

$$(1-\delta) + \delta\frac{Y}{x_0} > (2-\theta)\frac{g'(\theta)}{g(\theta)} \quad \forall\theta\in[\theta_L,\bar{\theta}].$$

As $x_0 < Y$*, a sufficiency condition is that*

$$\sup_{\theta\in\Theta}(2-\theta)\frac{g'(\theta)}{g(\theta)} \leq 1$$

4.3 Initial Debt and Implementation

We now proceed to the second step and find the optimal value of x_0 by solving the following problem:

$$\begin{aligned} &\max_{x_0}\ W_0(x_0) \\ &\text{subject to} \\ &x_0 q(x_0) \geq I. \end{aligned} \tag{28}$$

The constraint $x_0q(x_0) \geq I$ is both sufficient and necessary for the feasible set in problem (21) to be non-empty. It is easily established that this constraint holds if and only if $x_0 \in [x_0^{\min}, x_0^{\max}]$, where $x_0^{\min} < x_0^{\max}$ are the two roots to the equation $x_0q(x_0) = I$.[9]

Figure 4 illustrates the problem in (28). It builds on the left-hand boundary of the feasible set shown in Figure 3 and adds the borrower's indifference curves to represent her preferences, with darker curves indicating higher payoff. For a given θ_L (and thus a fixed debt limit), increasing x_0 always lowers the borrower's payoff. By contrast, for a given x_0, the borrower's payoff is maximized when θ_L (and hence the debt limit) is neither too high nor too low. A high debt limit offers excessive financial flexibility and leads to over-borrowing, whereas a low limit restricts flexibility and results in under-borrowing.

The next proposition characterizes the optimal initial borrowing and its associated debt limit.

Proposition 6 (Necessary Conditions Initial Debt)**.** *Let* x_0^{n*} *be a solution to* (28)*, and* $x_1^f(\theta_L^{n*})$ *the corresponding debt limit that solves* (21)*.*

1. *If* $x_1^*(\mathbb{E}[\theta]) \leq x_0^{\min}$ *there is no flexibility for additional financing at* $t = 1$*. In this case,* $x_0^{n*} = x_0^{\min}$ *and* $\theta_L^{n*} = \bar{\theta} = 1$*.*

[9] Assumption 2 guarantees the existence of the root. The fact $x_0q(x_0)$ is concave in x_0 implies that there are two roots.

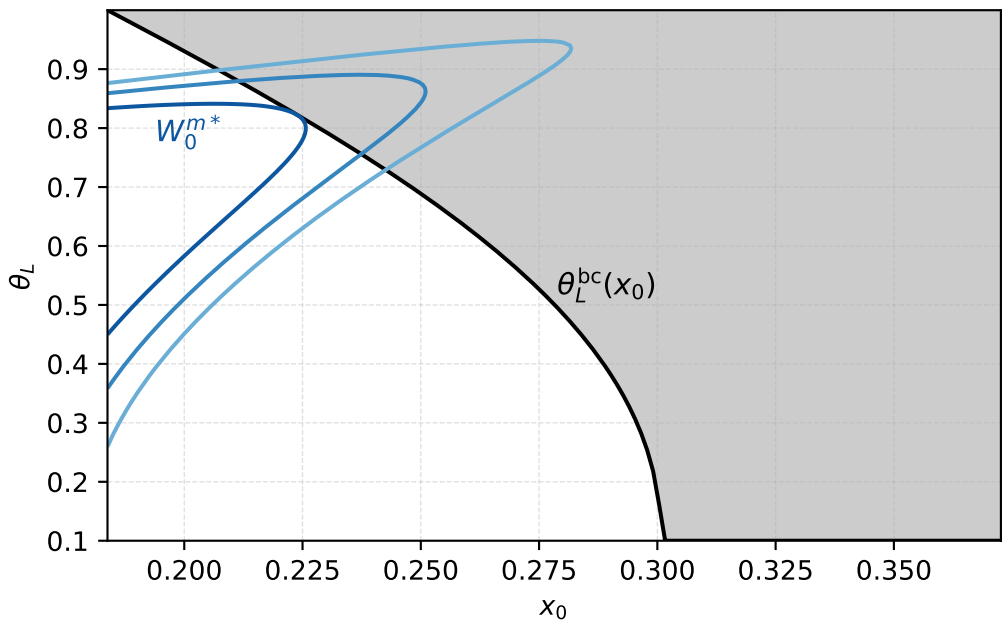

Figure 4: Indifference curves and determination of optimal pair (x_0, θ_L). We assume that $\theta \sim U[0.1, 1]$, $Y = 1$ and $c(q) = q^2$. The maximum pledgeable income is $I = 0.6 \times I^{\max}$ where $I^{\max} = \max_{x_1 \in [0,Y]} \; x_1 q(x_1)$. W_0^{n*} corresponds to the value of the optimization problem in (28).

2. *If $x_1^*(\mathbb{E}[\theta]) > x_0^{\min}$ there is flexibility for additional financing at $t = 1$. In this case, $\theta_L^{n*} < \bar{\theta} = 1$ and $x_0^{n*} > x_0^{\min}$. Specifically, $\{x_0^{n*}, \theta_L^{n*}\}$ jointly solve*

$$x_0^{n*}\mathbb{E}\left[q(x_1^f(\theta_L^{n*}, x_0^{n*}))\mathbf{1}_{\{\theta \le \theta_L^{n*}\}} + q(x_1^f(\theta, x_0^{n*}))\mathbf{1}_{\{\theta > \theta_L^{n*}\}}\right] = I$$

$$x_0^{n*} = \frac{(1-\delta)\mathbb{E}\left[q(x_1^f(\theta_L^{n*}, x_0^{n*}))\mathbf{1}_{\{\theta \le \theta_L^{n*}\}} + q(x_1^f(\theta_L^{n*}, x_0^{n*}))\mathbf{1}_{\{\theta > \theta_L^{n*}\}}\right]}{-\mathbb{E}\left[q'(x_1^f(\theta, x_0^{n*}))\frac{\partial x_1^f(\theta, x_0^{n*})}{\partial x_0}\mathbf{1}_{\{\theta > \theta_L^{n*}\}}\right]}$$

and

$$\delta = -\frac{x_0^{n*}}{\mathbb{E}[\theta_L^{n*} - \theta | \theta \le \theta_L^{n*}]} \frac{q'(x_1^f(\theta_L^{n*}, x_0^{n*}))}{q(x_1^f(\theta_L^{n*}, x_0^{n*}))}.$$

Proposition 6 includes two cases based on the comparison between $x_1^*(\mathbb{E}[\theta])$ and $x_0^{\min}$. In the first case, $x_1^*(\mathbb{E}[\theta]) < x_0^{\min}$, so optimal total debt without knowing θ falls below the minimum initial debt required to satisfy the financing constraint. Therefore, the financing constraint by itself already causes over-borrowing, and the borrower's incentive to dilute old debt at $t = 1$ further exacerbates the problem. The optimal policy therefore imposes a debt limit that eliminates any

further incentives of over-borrowing, which leads to no financial flexibility. In the second case, $x_1^*(\mathbb{E}[\theta]) \geq x_0^{\min}$, so the minimum initial debt remains below optimal total debt without knowing θ. The optimal policy preserves some financial flexibility to balance between over- and under-borrowing. At $t = 1$, the agent can still borrow flexibly when the liquidity shock is low (high θ).

A key feature of Proposition 6 is that the financing constraint always binds under non-exclusive lending. In other words, it is never optimal to borrow more than I at $t = 0$. Instead, the optimal strategy selects initial debt x_0 and debt limit θ_L that provide exactly enough capital to fund investment I and nothing more. The reason is straightforward: less borrowing at $t = 0$ reduces over-borrowing incentives at $t = 1$, creating greater financial flexibility.

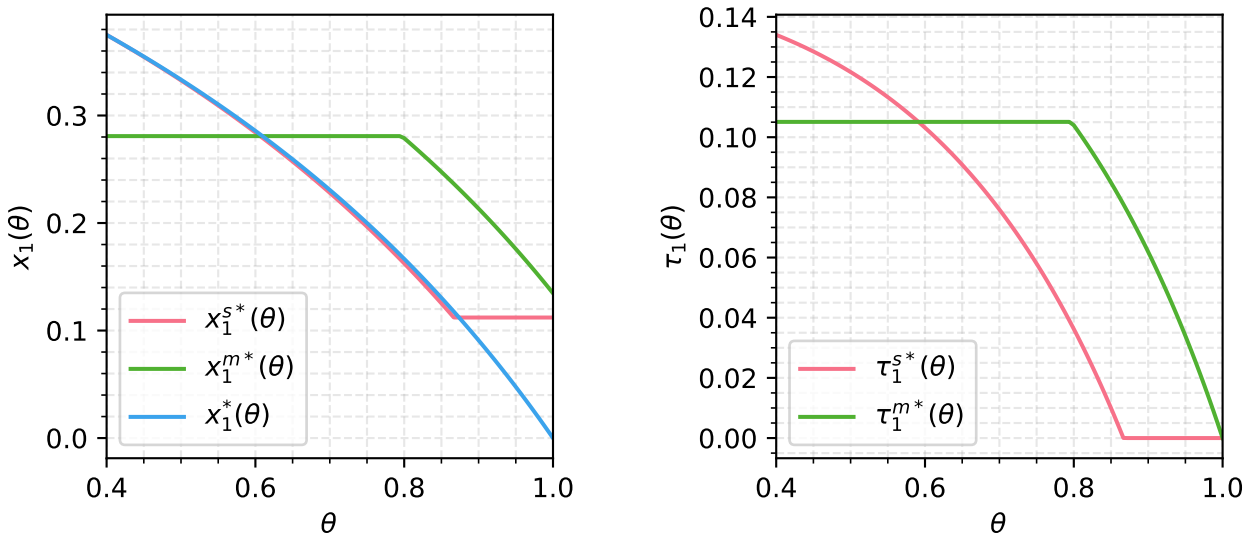

Figure 5: Optimal contract single lender and multiple lender. We assume that $\theta \sim U[0.4, 1]$, $Y = 1$, $c(q) = q^2$ and $I = 0.4 \times I^{\max}$, where the maximum pledgeable income is $I^{\max} = \max_{x_1 \in [0,Y]} \; x_1 q(x_1)$.

Figure 5 compares the optimal contract under multiple lenders, $x_1^{n*}(\theta)$, with the single-lender solution $x_1^{s*}(\theta)$ and efficient debt level $x_1^*(\theta)$. In the left panel, which plots total debt, $x_1^{n*}(\theta)$ is flat for low θ and decreases monotonically once θ exceeds θ_L. At low θ, the borrower underborrows relative to both the efficient debt level and single-lender solutions; at high θ, she overborrows. Under the given parameters, the financing constraint binds under the single-lender case (i.e., $I > I^s$). As a result, $x_1^{s*}(\theta)$ exceeds $x_1^*(\theta)$ for $\theta < \theta_H$, becoming flat for $\theta > \theta_H$, and eventually falling below $x_1^*(\theta)$. The right panel shows the transfer the borrower receives at $t = 1$. In the ex post-efficient allocation, transfers are normalized to zero. Under multiple lenders, transfers are constant for low

θ and decline with θ thereafter. They fall below the single-lender case when θ is low but exceed it when θ is high.

Figure 6 shows how the optimal debt policy under multiple lenders varies with investment size I. As I increases, the initial debt level x_0 also rises, amplifying the borrower's incentive to dilute initial lenders at $t = 1$. This exacerbates over-borrowing, requiring a tighter debt limit. As a result, a larger range of types faces binding constraints ex-post. However, for types that remain unconstrained, the debt level increases with I.

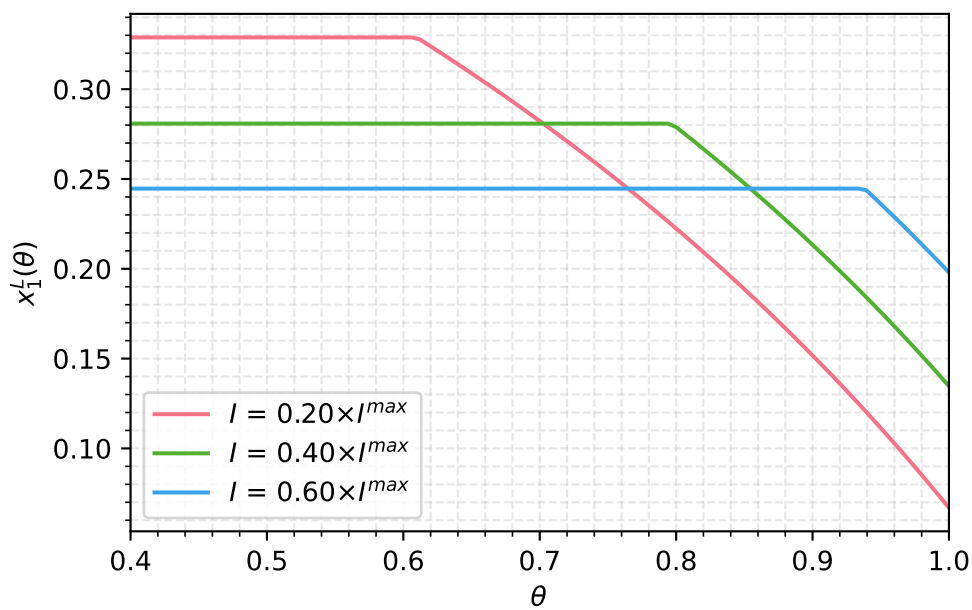

Figure 6: Comparative statics of face value at $t = 1$ for changes in initial investment I. We assume that $\theta \sim U[0.4, 1]$, $Y = 1$ and $c(q) = q^2$.

The solution identified by Proposition 6 can be implemented using a widely observed contractual mechanism: an initial debt issuance x_0^{n*} paired with a maintenance covenant that restricts total indebtedness to not exceed $x_1^f(\theta_L^{n*}, x_0^{n*})$. Such maintenance covenants are commonplace in corporate bond issuances and credit agreements. This arrangement restricts financial flexibility, as the borrower at $t = 1$ sometimes finds herself constrained from accessing her preferred level of additional debt when faced with financing needs. Meanwhile, this covenant creates protection for existing creditors by restricting the dilution problem.

Linear Example. *Let*

$$\hat{x}_0 \equiv \frac{x_0}{Y}, \qquad \hat{x}_1 \equiv \frac{x_1}{Y}, \qquad \kappa \equiv \frac{cI}{Y^2}.$$

The minimum feasible borrowing at $t = 0$ is

$$\hat{x}_0^{\min} = \frac{1}{2}\left(1 - \sqrt{1 - 4\kappa}\right)$$

and

$$\hat{x}_1^*(\mathbb{E}[\theta]) = \frac{1 - \mathbb{E}[\theta]}{2 - \mathbb{E}[\theta]}.$$

From here, we get that $\hat{x}_1^*(\mathbb{E}[\theta]) > \hat{x}_0^{\min}$ if

$$\mathbb{E}[\theta] < \frac{2\sqrt{1 - 4\kappa}}{1 + \sqrt{1 - 4\kappa}}.$$

In this case, $(\hat{x}_0, \theta_L)$ solve

$$\kappa = \hat{x}_0(1 - \hat{x}_0)\mathbb{E}\left[\frac{1}{2 - \theta_L}\mathbf{1}_{\{\theta \le \theta_L\}} + \frac{1}{2 - \theta}\mathbf{1}_{\{\theta > \theta_L\}}\right]$$

$$\frac{\hat{x}_0}{1 - \hat{x}_0} = \frac{(1 - \delta)\mathbb{E}\left[\frac{1}{2-\theta_L}\mathbf{1}_{\{\theta \le \theta_L\}} + \frac{1}{2-\theta}\mathbf{1}_{\{\theta > \theta_L\}}\right]}{\mathbb{E}\left[\frac{1}{2-\theta}\mathbf{1}_{\{\theta > \theta_L\}}\right]}$$

where

$$\delta = \frac{2 - \theta_L}{\mathbb{E}[\theta_L - \theta | \theta \le \theta_L]}\frac{\hat{x}_0}{1 - \hat{x}_0}.$$

Comparative Statics. In the linear example, after appropriately scaling by Y, the solution to the problem depends only on the parameter κ and the distribution of liquidity shocks. The following proposition provides comparative statics with respect to changes in κ and in the liquidity-shock distribution. For the latter, we consider a family of distributions $\{g(\cdot;\phi)\}_{\phi \in [\underline{\phi}, \overline{\phi}]}$ on a common support $\Theta = [\underline{\theta}, 1]$ that satisfies the log-concavity assumption 1 and is ordered according to the Monotone Likelihood Ratio Order (MLR), so that the density $g(\theta;\phi)$ is log-supermodular in (θ, ϕ).

Proposition 7. *Consider the linear example and define $\hat{x}_0 \equiv x_0/Y$. For each pair (κ, ϕ), let $\theta_L^*(\kappa, \phi)$ and $\hat{x}_0^*(\kappa, \phi)$ denote the solution to the optimality conditions in Proposition 6, and let $\mathbb{E}_\phi[\cdot]$ denote expectation under $g(\cdot;\phi)$. Suppose that*

$$x_1^*(\mathbb{E}_\phi[\theta]) > x_0^{\min},$$

for every (κ, ϕ) under comparison. Assume also that the family introduced above is ordered by MLR and that each $G(\cdot;\phi)$ is log-concave. Then:

1. *$\theta_L^*(\kappa, \phi)$ is nondecreasing in κ and in ϕ.*

2. $\hat{x}_0^*(\kappa, \phi)$ is nondecreasing in κ and nonincreasing in ϕ.

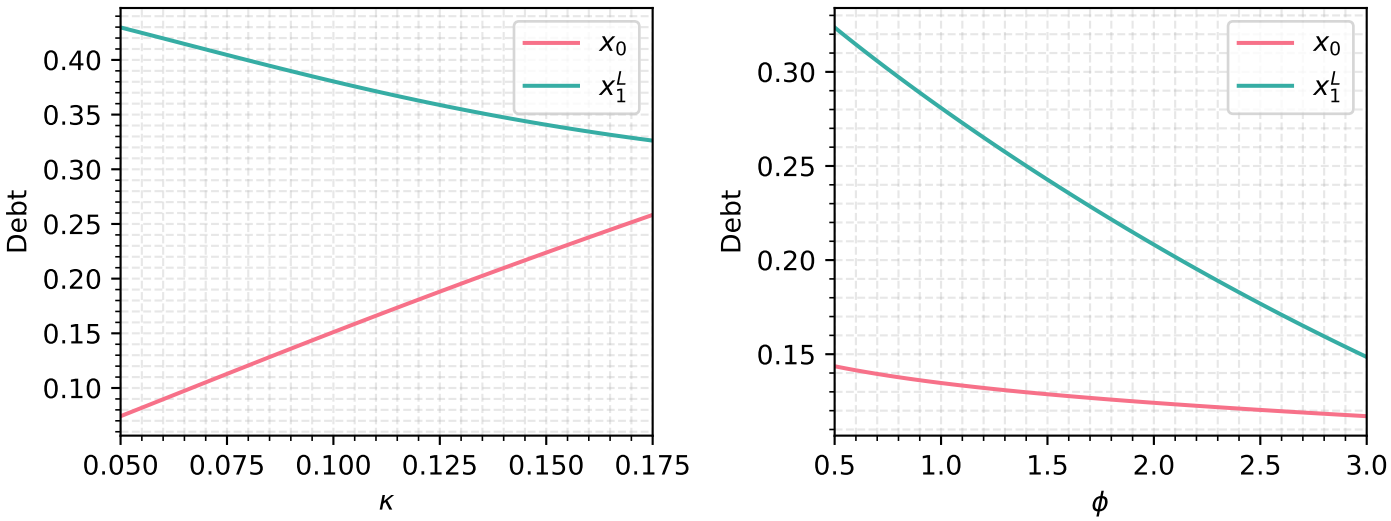

Figure 7: Comparative statics of optimal debt under multiple lenders. We assume $Y = 1$, $c(q) = q^2$, and $G(\theta) = \left(\frac{\theta - \underline{\theta}}{1 - \underline{\theta}}\right)^{\phi}$ with $\underline{\theta} = 0.4$. In the left panel, we fix $\phi = 1$ and vary $\kappa \in [0.05, 0.175]$. In the right panel, we fix $\kappa = 0.1$ and vary $\phi \in [0.5, 3]$.

We plot these results in Figure 7. The left panel shows how the optimal solution under multiple lenders varies with κ, the scaled financing requirement in the linear example. As κ increases, the borrower must raise more initial funds, leading to a higher x_0^{m*}. To mitigate stronger incentives to overborrow at $t = 1$, the optimal debt limit x_1^{L*} decreases. The right panel shows how the multiple-lender solution varies with the distribution parameter ϕ in the power family

$$G(\theta; \phi) = \left(\frac{\theta - \underline{\theta}}{1 - \underline{\theta}}\right)^{\phi}.$$

This family is ordered by the monotone likelihood ratio order, so a higher ϕ shifts probability mass toward larger values of θ. Equivalently, the mean

$$\mathbb{E}_{\phi}[\theta] = \underline{\theta} + \frac{\phi}{\phi + 1}(1 - \underline{\theta})$$

is increasing in ϕ. As ϕ increases, there is less need to borrow at $t = 1$, and the value of financial flexibility becomes less important. As a result, overborrowing becomes a greater concern. Consequently, the optimal debt limit x_1^{L*} decreases to provide tighter borrowing constraints. Since

initial lenders face lower dilution, they are willing to lend more for any given x_0. To raise the required investment, the optimal initial debt x_0^{m*} can therefore be lower. These comparative statics describe the interior regime with $\theta_L^* < \bar{\theta}$, in which the borrower retains some financial flexibility at $t = 1$. Once parameters cross into the no-flexibility region, the solution becomes $\theta_L^* = \bar{\theta} = 1$, so $x_1^{L*} = x_0^{m*} = x_0^{\min}$. In that regime, changes in κ operate through the minimum feasible initial debt, not through an interior adjustment in the optimal debt limit.

Next, we examine the effect of uncertainty by examining the impact of increasing the dispersion in the distribution of liquidity shocks θ. In the following example, we model the liquidity shocks θ using a beta distribution, because this allows us to increase the dispersion while keeping a constant mean without changing the support $[\underline{\theta}, \bar{\theta}]$. By varying the shape parameters appropriately, we can adjust the dispersion of θ without altering its mean or its bounded support. In particular, we set $\bar{\theta} = 1$ and let

$$g(\theta) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \left(\frac{\theta - \underline{\theta}}{1 - \underline{\theta}}\right)^{\alpha-1} \left(\frac{1-\theta}{1-\underline{\theta}}\right)^{\beta-1} \frac{1}{1-\underline{\theta}}, \ \forall \theta \in [\underline{\theta}, 1],$$

where $\Gamma(\cdot)$ denotes the gamma function and

$$\alpha = k\frac{\mathbb{E}[\theta] - \underline{\theta}}{1 - \underline{\theta}}$$
$$\beta = k\frac{1 - \mathbb{E}[\theta]}{1 - \underline{\theta}}$$

The spread factor k measures the dispersion of the distribution and is inversely related to the variance of the distribution. In particular, the standard deviation of θ is

$$\sigma(\theta) = \sqrt{\frac{(1 - \mathbb{E}[\theta])(\mathbb{E}[\theta] - \underline{\theta})}{k+1}}.$$

Figure 8 shows that the debt limit gets higher as σ increases because the value of financial flexibility increases with the uncertainty in liquidity shock.

5 Alternative Financing Arrangements

5.1 Short-Term Debt

To highlight the non-exclusivity problem, we have restricted the borrower to issuing only long-term debt due at $t = 2$. We now extend the analysis by introducing short-term debt. We emphasize, however, that debt maturity is not our paper's central focus.

Specifically, we allow the borrower to finance the upfront investment at $t = 0$ using a combi-

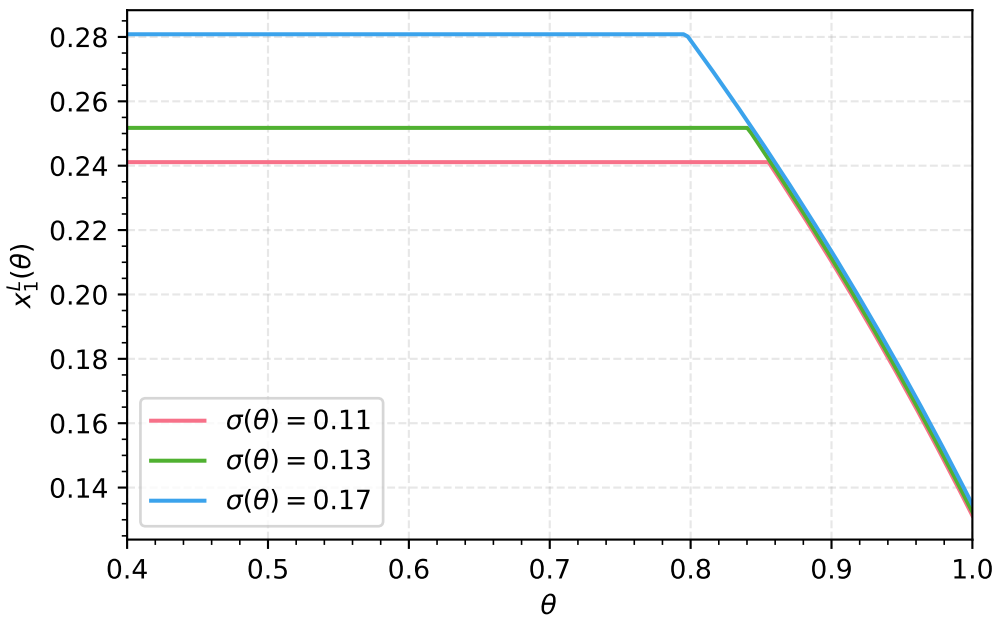

Figure 8: Effect of mean-preserving spread in liquidity shocks. We assume $Y = 1$, $c(q) = q^2$, and $G(\theta)$ is beta distributed with $\mathbb{E}[\theta] = 0.7$ and $\Theta = [0.4, 1]$.

nation of short-term debt b that matures at $t = 1$, and long-term debt x_0 that matures at $t = 2$. Short-term debt is due after the liquidity shock θ is realized. Because the borrower is financially constrained, she must issue new debt to repay the short-term debt b due at that time.

It becomes immediately clear that short-term debt must be riskless because there is no cash-flow risk at $t = 1$.[10] Our next result shows that short-term debt completely crowds out long-term debt.

Proposition 8. *If the borrower can issue both short- and long-term debt, she will exclusively borrow short-term debt $b = I$ due at $t = 1$. At $t = 1$, she rolls over the short-term debt by borrowing $\max\{x_1^*(\theta), x_1^b\}$ from new lenders, where x_1^b satisfies $b = x_1^b q(x_1^b)$.*

Why does short-term debt completely crowd out long-term debt? As we show in Section 4, any long-term debt issued at $t = 0$ gives the borrower incentives to dilute it at $t = 1$, exacerbating the overborrowing problem. By contrast, short-term debt avoids this dilution and the associated overborrowing because it matures before the borrower can issue again. Therefore, it is optimal at $t = 0$ to finance the entire investment I with short-term debt. At the refinancing date $t = 1$, the

[10] A proof goes as follows. If the borrower were to default under some report $\hat{\theta}$ (resulting in zero payoff), she would instead choose a report $\hat{\theta}'$ that avoids default and yields a positive payoff. Thus, truthful reporting requires that default never occur, so that short-term debt must be riskless.

borrower has no legacy debt and prefers to issue the efficient debt $x_1^*(\theta)$. If this is sufficient to repay $b = I$, the borrower consumes the remaining proceeds. However, if this level of debt is insufficient to repay $b = I$, the borrower must issue a larger amount x_1^b where $x_1^b q(x_1^b) = b$ to meet the rollover requirement.

Our finding that short-term debt completely crowds out long-term debt should be interpreted cautiously, because the analysis abstracts away several downsides of short-term debt. Due to these omitted costs and risks, firms rarely issue only short-term debt in practice. For instance, our model assumes frictionless debt markets, thereby ignoring transaction costs associated with rolling over short-term debt. Moreover, our framework generates no cash flows at $t = 1$, and results would differ under uncertain intermediate cash flows. In a related setting with such uncertainty, Hu et al. (2021) show that borrowers issue long-term debt when short-term debt becomes risky.

While short-term debt crowds out long-term debt, it still fails to implement the constrained optimal allocation. This occurs because repayment at $t = 1$ is not contingent on the borrower's liquidity shock. The efficient debt level allows the borrower to pledge more cash flow when θ is low and less when θ is high. In the next subsection, we show how long-term debt with acceleration clauses can achieve this allocation.

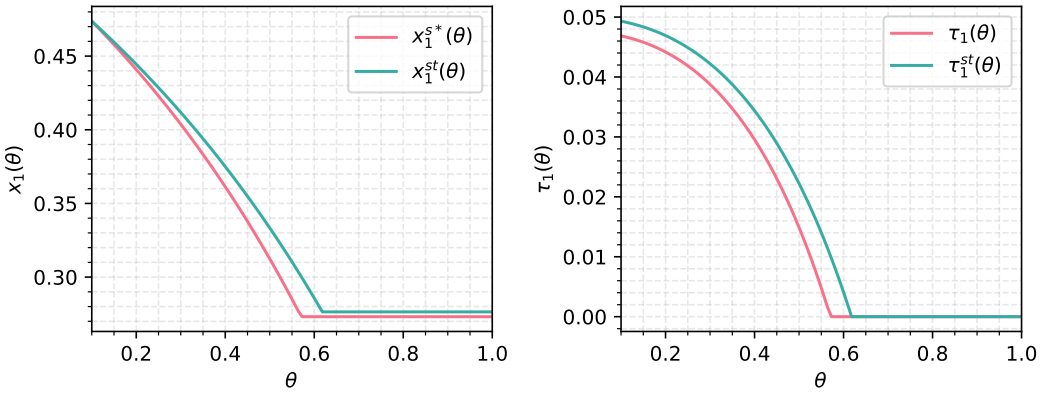

Figure 9: Optimal contract with short-term debt compared to the single lender contract. We assume that $\theta \sim U[0.1, 1]$, $Y = 1$, $c(q) = q^2$, and $I = 0.8 \times I^{\max}$.

5.2 Contingent Prepayment

Until now, we have focused on mechanisms that restrict the borrower from issuing additional debt, traditional restrictive covenants common in loan agreements. Instead of outright prohibiting new debt, a contract could require prepayment conditional on the borrower's actions. Under a

standard restrictive covenant, a violation leads to default and acceleration of the entire debt. By contrast, a covenant could stipulate that certain borrower actions trigger partial or full prepayment of principal without constituting default.

This type of contingent prepayment is called a *sweep*. Loan agreements often include several forms of sweeps. An *equity sweep* requires that a fixed share of the proceeds from new equity issuance be used to repay existing debt. An *asset sweep* directs part of the proceeds from asset sales to debt repayment. Most relevant here, a *debt sweep* requires that a portion of the funds raised through new debt issuance be used to prepay outstanding loans. Importantly, a sweep does not necessarily apply to the full amount: a new debt issue might obligate the borrower to repay only part of the existing loan rather than the entirety. In this way, sweep provisions mitigate agency conflicts while still leaving borrowers flexibility to raise additional funds.

Although less common than traditional covenants, sweep provisions are widely used in practice. According to Bratton (2016), sweeps became prominent features of bank loan contracts during the 1990s. Bradley and Roberts (2015) document this rise in their review of loan agreements: in 1993, 32% of loans included an asset sweep, 18% a debt sweep, and 24% an equity sweep. By 2001, these figures had risen to 93%, 80%, and 75%, respectively.[11] Similarly, Eckbo et al. (2023) finds that debtor-in-possession (DIP) loans and leveraged loans frequently include mandatory prepayment terms, almost always featuring a debt issuance sweep. In such cases, proceeds from any new debt must first be used to pay down existing loans.[12]

Next, we extend our model to analyze such contingent prepayment policies. Our main result will be to show that a properly designed debt sweep can implement the single lender allocation, thus completely eliminating the cost of non-exclusivity. Suppose when the borrower reports $\hat{\theta}$, an amount $b(\hat{\theta}) \in [0, x_0]$ becomes due at $t = 1$. Under truthful reporting ($\hat{\theta} = \theta$), let

$$x_1^+(\theta) \equiv x_1(\theta) - b(\theta)$$

be the face value of debt after prepayment and

$$m_1(\theta) \equiv b(\theta)\left(1 - q(x_1^+(\theta))\right)$$

be the effective transfer to the initial lenders. At $t = 1$, the net proceeds from debt issuance and

[11] Bratton (2016) note that the trend toward looser covenants in more recent years has reduced the use of sweep provisions.

[12] While the most common arrangement requires that 100% of proceeds be applied to prepayment, contracts often required that only a fraction of the proceeds go to prepayment. For example, in a loan received by Avon Products on June 29, 2012, the credit agreement required the company to prepay fifty percent (50%) of the net cash proceeds from any debt issuance exceeding $500 million. (Avon Products, Inc., 2014)

mandatory repayments are:

$$\tau_1(\theta) = (x_1^+(\theta) - x_0)q(x_1^+(\theta)) - m_1(\theta),$$

and the initial lenders' participation constraint at $t = 0$ becomes

$$\tau_0 = \mathbb{E}\left[x_0 q(x_1^+(\theta)) + m_1(\theta)\right].$$

Substituting both expressions into the borrower's $t = 0$ expected payoff, it is easily derived that the borrower aims to maximize $\mathbb{E}\left[W_1(x_1^+(\theta), \theta)\right] - I$. Therefore, We can write the borrower's $t = 0$ problem under mandatory prepayment as follows:

$$\begin{aligned}
&\max_{x_0, x_1^+(\theta), m_1(\theta) \geq 0} \mathbb{E}\left[W_1(x_1^+(\theta), \theta)\right] \\
&\text{subject to} \\
&\theta \in \arg\max_{\hat{\theta}} \left\{V_1(x_1^+(\hat{\theta}), x_0, \theta) - m_1(\hat{\theta})\right\} \\
&\mathbb{E}\left[x_0 q(x_1^+(\theta)) + m_1(\theta)\right] \geq I \\
&m_1(\theta) \leq (x_1^+(\theta) - x_0)q(x_1^+(\theta)) \\
&m_1(\theta) \leq x_0\left(1 - q(x_1^+(\theta))\right).
\end{aligned} \tag{29}$$

The first constraint is derived from truthful reporting, the second requires $\tau_0 \geq I$ for financing, and the last two follow from $\tau_1(\theta) \geq 0$ and $b(\theta) \leq x_0$, respectively.

We have the following result.

Proposition 9. *The solution to problem (29) is $x_0 = x_1^{s*}(\bar{\theta})$, $x_1^{+*}(\theta) = x_1^{s*}(\theta)$ together with*

$$m_1(\theta) = V_1(x_1^{s*}(\theta), x_1^{s*}(\bar{\theta}), \theta) - \theta_H V_2(x_1^{s*}(\theta_H)) + \int_\theta^{\theta_H} V_2(x_1^{s*}(\tilde{\theta}))d\tilde{\theta}, \tag{30}$$

where $x_1^{s*}(\theta)$ is the solution to the single lender problem in Proposition 3.

Proposition 9 shows that the ultimate total debt level determined at $t = 1$ is identical to the single-lender outcome. As a result, under lending non-exclusivity with mandatory prepayment, the borrower receives the same payoff as if she could borrow exclusively. This holds because mandatory prepayment allows for state-contingent transfers to initial lenders. While initial lenders still anticipate dilution at $t = 1$, they get compensated by the effective transfers that depend on the realized liquidity shock. Note that Proposition 9 also shows that $x_0 = x_1^{s*}(\bar{\theta})$, the optimal debt level for the borrower with the lowest liquidity shock. This follows from the limited liability

constraint $\tau_1(\theta) \geq 0$, which becomes binding if $\theta = \bar{\theta}$.

Our next result discusses the implementation of the mandatory prepayment provision via a debt issuance sweep.

Proposition 10 (Debt Sweep Implementation)**.** *Let*

$$\tilde{x}_1(\theta) \;=\; x_1^{s*}(\theta) + b(\theta),$$

denote the total debt at $t = 1$ without mandatory prepayment, where

$$b(\theta) \;=\; \frac{m_1(\theta)}{1 - q(x_1^{s*}(\theta))},$$

*and $m_1(\theta)$ in equation (30). Define $\theta(\cdot) \equiv \tilde{x}_1^{-1}(\cdot)$ as the inverse of $\tilde{x}_1(\theta)$. The mandatory prepayment described in Proposition 9 can be implemented as a **debt issuance sweep**, in which a fraction*

$$\alpha(\Delta x_1) \;=\; \frac{b(\theta(x_0 + \Delta x_1))}{q\Big(x_0 + \Delta x_1 - b(\theta(x_0 + \Delta x_1))\Big)\,\Delta x_1}.$$

of the proceeds from new debt issuance are used to prepay existing debt.

Figure 10 illustrates the debt sweep in the linear example.

6 Robustness and Extension

6.1 Alternative Specification Liquidity Shock

An alternative approach to modeling liquidity shocks involves the project's need for a specific amount of liquidity at the interim date. Following Holmström and Tirole (1998), we modify the model as follows.

The borrower has no time discount between $t = 1$ and $t = 2$. Instead, she needs to inject an additional amount of liquidity ηI at $t = 1$, where $\eta \in [0, \bar{\eta}]$ follows the distribution $G(\cdot)$ with density $g(\cdot)$ and is privately known to the borrower.

We first derive the borrower's desired amount of total debt in the absence of liquidity needs. By setting $\theta = 1$ in equation (24), we obtain $x_1^f = x_0$. Therefore, in the absence of liquidity needs, the borrower has no incentive to borrow, because there are no gains from debt issuance at $t = 1$. This suggests that borrowing is only needed to cover the liquidity needs. The solution in this case, therefore, closely follows Holmström and Tirole (1998): there exists a threshold η^* such that the

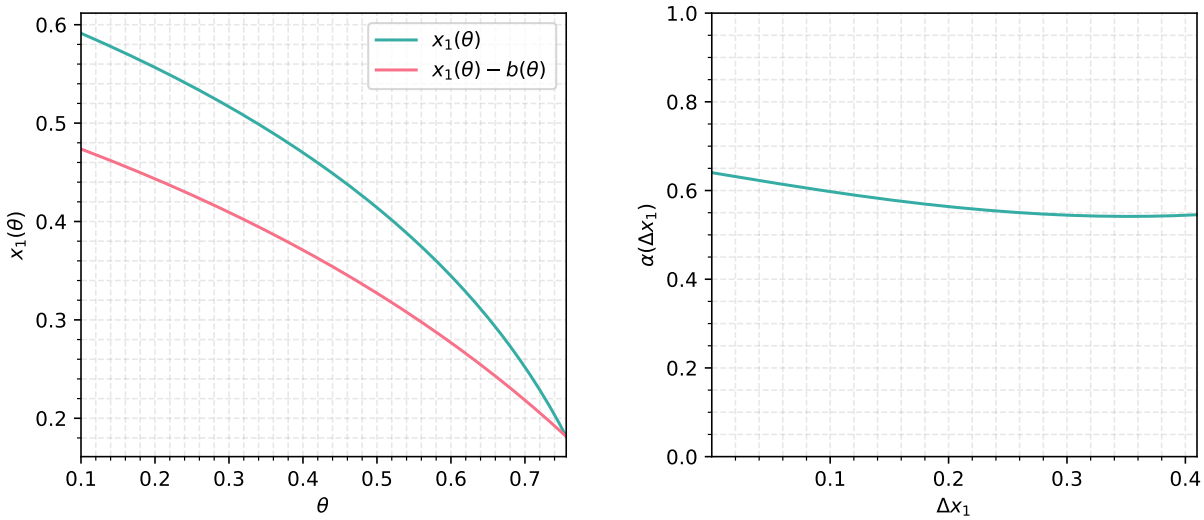

Figure 10: Implementation of optimal contract using non-linear debt sweep. We assume that $\theta \sim U[0.4, 1]$, $Y = 1$, $c(q) = q^2$, and $I = 0.4 \times I^{\max}$.

borrower's liquidity needs are fully satisfied when $\eta \leq \eta^*$, but not when $\eta > \eta^*$. This threshold η^* corresponds to a limit on the total level of debt.

If, in addition, we assume that the borrower applies a discount rate $\theta < 1$ between $t = 1$ and $t = 2$, where θ is common knowledge, this introduces incentives to over-borrow at $t = 1$ due to the well-documented dilution effect. In this case, the optimal policy must balance the incentives to over-borrow against the liquidity need.

Overall, modeling liquidity shocks as an amount of liquidity needed at the interim date is less interesting because the incentives to over-borrow are independent of the liquidity shock η. Therefore, we choose to model liquidity shock as one affecting the discount rate, as presented in our baseline model.

7 Conclusion

This paper examines the optimal design of financial flexibility for firms facing financing constraints and non-exclusive lending. We show that the tradeoff between preserving access to liquidity and preventing excessive borrowing leads to an endogenous debt limit as the optimal contractual solution. This limit allows firms to borrow flexibly when facing modest liquidity shocks while constraining over-borrowing during severe shocks. Our model provides a theoretical foundation for the

prevalence of maintenance covenants in debt contracts, demonstrating how they mitigate creditor dilution while preserving necessary financial flexibility. The analysis highlights how financing constraints shape firms' borrowing capacity and offers insights into the trade-offs involved in designing optimal debt contracts under asymmetric information and non-exclusive lending. Future research could explore how stochastic cash flows or alternative covenant structures might further refine these contractual solutions.

References

Admati, A. R., P. M. Demarzo, M. F. Hellwig, and P. Pfleiderer (2018). The leverage ratchet effect. Journal of Finance 73(1), 145–198.

Aghion, P. and P. Bolton (1992). An incomplete contracts approach to financial contracting. The review of economic Studies 59(3), 473–494.

Aguiar, M., M. Amador, H. Hopenhayn, and I. Werning (2019). Take the short route: Equilibrium default and debt maturity. Econometrica 87(2), 423–462.

Amador, M. and K. Bagwell (2013). The theory of optimal delegation with an application to tariff caps. Econometrica 81(4), 1541–1599.

Amador, M., K. Bagwell, and A. Frankel (2018, October). A note on interval delegation. Economic Theory Bulletin 6(2), 239–249.

Amador, M., I. Werning, and G. M. Angeletos (2006). Commitment vs. flexibility. Econometrica 74(2), 365–396.

Avon Products, Inc. (2014). Form 10-K, Annual Report for the fiscal year ended December 31, 2013. https://www.sec.gov/Archives/edgar/data/8868/000000886814000013/avp10k2013.htm. U.S. Securities and Exchange Commission; accessed August 25, 2025.

Bagnoli, M. and T. Bergstrom (2005). Log-Concave Probability and Its Applications. Economic Theory 26(2), 445–469.

Baron, D. P. and D. Besanko (1984). Regulation and information in a continuing relationship. Information Economics and policy 1(3), 267–302.

Black, F. and M. Scholes (1973). The pricing of options and corporate liabilities. Journal of Political Economy 81(3), 637–654.

Bradley, M. and M. R. Roberts (2015, June). The Structure and Pricing of Corporate Debt Covenants. Quarterly Journal of Finance 05(02), 1550001.

Bratton, W. W. (2016, October). Bond and loan covenants, theory and practice. Capital Markets Law Journal 11(4), 461–485.

DeMarzo, P. M. and Z. He (2021). Leverage dynamics without commitment. Journal of Finance 76(3), 1195–1250.

Eckbo, B. E., K. Li, and W. Wang (2023). Loans to chapter 11 firms: contract design, repayment risk, and pricing. The Journal of Law and Economics 66(3), 465–509.

Fama, E. F. and M. Miller (1972). The Theory of Finance. Holt Rinehart & Winston.

Gamba, A. and A. Triantis (2008). The value of financial flexibility. The journal of finance 63(5), 2263–2296.

Graham, J. R. (2022). Presidential address: Corporate finance and reality. The Journal of Finance 77(4), 1975–2049.

Halac, M. and P. Yared (2014). Fiscal rules and discretion under persistent shocks. Econometrica 82(5), 1557–1614.

Halac, M. and P. Yared (2022). Fiscal rules and discretion under limited enforcement. Econometrica 90(5), 2093–2127.

Holmström, B. and J. Tirole (1998). Private and public supply of liquidity. Journal of political Economy 106(1), 1–40.

Hu, Y., F. Varas, and C. Ying (2021). Debt maturity management. Technical report, Working paper.

Jensen, M. C. (1986). Agency costs of free cash flow, corporate finance, and takeovers. The American economic review 76(2), 323–329.

Leland, H. E. (1994). Corporate debt value, bond covenants, and optimal capital structure. The journal of finance 49(4), 1213–1252.

Luenberger, D. G. (1969). Optimization by Vector Space Methods. John Wiley & Sons.

Milgrom, B. Y. P. and C. Shannon (1994). Monotone comparative statics. Econometrica 62(1), 157–180.

Shaked, M. and J. G. Shanthikumar (2007). Stochastic Orders. Springer Series in Statistics. New York: Springer.

Simpson Thacher & Bartlett LLP (2022). Leveraged finance 101: A covenant handbook. Technical report, Simpson Thacher & Bartlett LLP. Available at https://www.stblaw.com/about-us/publications/view/2022/09/12/simpson-thacher-publishes-em-leveraged-finance-101-a-covenant-handbook-em.

Appendix

A Single Lender Equilibrium

For notational convenience, let us define

$$\bar{V}_1 = \tau_1(\bar{\theta}) + \bar{\theta} V_2(x_1(\bar{\theta}))$$

as type $\bar{\theta}$'s continuation payoff at $t = 1$.

Lemma 3. *Any incentive-compatible pair* $\{\tau_1(\theta), x_1(\theta)\}$ *satisfying* $\bar{V}_1 \geq \bar{\theta} V_2(x_1(\bar{\theta}))$ *satisfies the constraint*

$$\tau_1(\theta) = \bar{V}_1 - \theta V_2\left(x_1(\theta)\right) - \int_{\theta}^{\bar{\theta}} V_2(x_1(\tilde{\theta})) d\tilde{\theta} \geq 0 \quad \forall \theta \in \Theta.$$

Proof of Lemma 3

Proof. We verify that it is sufficient to consider the non-negativity constraint of $\tau_1(\theta)$ for the highest type. Integration by parts implies that

$$\theta V_2\left(x_1(\theta)\right) + \int_{\theta}^{\bar{\theta}} V_2(x_1(\tilde{\theta})) d\tilde{\theta} = \bar{\theta} V_2\left(x_1(\bar{\theta})\right) - \int_{\theta}^{\bar{\theta}} \tilde{\theta} V_2'(x_1(\tilde{\theta})) dx_1(\tilde{\theta})$$

Hence, we can write the envelope condition as

$$\tau_1(\theta) = \tau_1(\bar{\theta}) + \int_{\theta}^{\bar{\theta}} \tilde{\theta} V_2'(x_1(\tilde{\theta})) dx_1(\tilde{\theta})$$

It follows that, for any $\theta'' > \theta'$, we have

$$\tau_1(\theta'') = \tau_1(\theta') - \int_{\theta'}^{\theta''} \tilde{\theta} \underbrace{V_2'(x_1(\tilde{\theta}))}_{<0} \underbrace{dx_1(\tilde{\theta})}_{\leq 0} \leq \tau_1(\theta'),$$

which means that $\tau_1(\bar{\theta}) \geq 0$ implies $\tau_1(\theta) \geq 0$ for all $\theta < \bar{\theta}$. Moreover, the constraint $\tau(\bar{\theta}) \geq 0$ is equivalent to require that $\bar{V}_1 \geq \bar{\theta} V_2\left(x_1(\bar{\theta})\right)$ □

Proof of Proposition 2

Proof. Let $\mathbf{x}_1 \equiv \{x_1(\theta)\}_{\theta\in\Theta}$. For a fixed λ (and therefore a fixed δ) we have

$$\max_{\mathbf{x}_1} \mathcal{U}(\mathbf{x}_1) \equiv \int_{\underline{\theta}}^{\bar{\theta}} \Big[W_1(x_1(\theta),\theta) + (1-\delta)\frac{G(\theta)}{g(\theta)}V_2\left(x_1(\theta)\right)\Big]dG(\theta) - (1-\delta)\bar{\theta}V_2\left(x_1(\bar{\theta})\right)$$

s.t. $\mathbf{x}_1$ is non-increasing.

Starting from $\mathbf{x}_1$, the directional derivative in the direction $h(\theta)$ is generally defined as

$$\nabla\mathcal{U}(\mathbf{x}_1;\mathbf{h}) = \frac{d}{dk}\mathcal{U}(\mathbf{x}+k\mathbf{h})\Big|_{k=0},$$

which, in our case, is given by

$$\nabla\mathcal{U}(\mathbf{x}_1;\mathbf{h}) = (1-\delta)\bar{\theta}q\left(x_1(\bar{\theta})\right)h(\bar{\theta}) + \int_{\underline{\theta}}^{\bar{\theta}} \Big[\frac{\partial W_1(x_1(\theta),\theta)}{\partial x_1}g(\theta) - (1-\delta)G(\theta)q\left(x_1(\theta)\right)\Big]h(\theta)d\theta.$$

Letting

$$\Phi(\theta;\mathbf{x}_1) \equiv \int_{\underline{\theta}}^{\theta} \Big[\frac{\partial W_1(x_1(\tilde{\theta}),\tilde{\theta})}{\partial x_1}g(\tilde{\theta}) - (1-\delta)G(\tilde{\theta})q\left(x_1(\tilde{\theta})\right)\Big]d\tilde{\theta},$$

the directional derivative can be written as

$$\nabla\mathcal{U}(\mathbf{x}_1;\mathbf{h}) = (1-\delta)\bar{\theta}q\left(x_1(\bar{\theta})\right)h(\bar{\theta}) + \int_{\underline{\theta}}^{\bar{\theta}} \Phi'(\theta;\mathbf{x}_1)h(\theta)d\theta.$$

Using integration by parts, we obtain that $\nabla\mathcal{U}(\mathbf{x}_1;\mathbf{h})$ can be written as

$$\nabla\mathcal{U}(\mathbf{x}_1;\mathbf{h}) = \left[\Phi(\bar{\theta};\mathbf{x}_1) + (1-\delta)\bar{\theta}q\left(x_1(\bar{\theta})\right)\right]h(\bar{\theta}) - \int_{\underline{\theta}}^{\bar{\theta}} \Phi(\theta;\mathbf{x}_1)dh(\theta).$$

This alternative formulation for the directional derivative is convenient because it is written explicitly in terms of $dh(\theta)$ rather than $h(\theta)$. This allows us to incorporate the monotonicity constraint directly into the analysis of the first-order conditions.

Let P be set of all non-increasing functions on Θ. Lemma 1 in (Luenberger, 1969, p. 227) provides the following necessary and sufficient condition for $\mathbf{x}_1$ to maximize $\mathcal{U}$:

$$\nabla\mathcal{U}(\mathbf{x}_1;\tilde{\mathbf{x}}_1) \leq 0 \quad \forall\ \tilde{\mathbf{x}}_1 \in P$$
$$\nabla\mathcal{U}(\mathbf{x}_1;\mathbf{x}_1) = 0.$$

In our case, these two conditions become

$$\left[\Phi(\bar{\theta};\mathbf{x}_1)+(1-\delta)\bar{\theta}q\left(x_1(\bar{\theta})\right)\right]\tilde{x}_1(\bar{\theta})-\int_{\underline{\theta}}^{\bar{\theta}}\Phi(\theta;\mathbf{x}_1)d\tilde{x}_1(\theta)\leq 0\quad\forall\ \tilde{\mathbf{x}}_1\in P \tag{31a}$$

$$\left[\Phi(\bar{\theta};\mathbf{x}_1)+(1-\delta)\bar{\theta}q\left(x_1(\bar{\theta})\right)\right]x_1(\bar{\theta})-\int_{\underline{\theta}}^{\bar{\theta}}\Phi(\theta;\mathbf{x}_1)dx_1(\theta)=0. \tag{31b}$$

We are going to construct a solution, $\mathbf{x}_1$, given by

$$x_1(\theta)=\begin{cases}x_1^s(\theta) & \text{if } \theta\in[\underline{\theta},\theta_H)\\ x_1^s(\theta_H) & \text{if } \theta\in[\theta_H,\bar{\theta}]\end{cases}$$

and show that it satisfies (31). Note (14) implies that $x_1^s(\theta)$ solves

$$\frac{\partial W_1(x_1,\theta)}{\partial x_1}-(1-\delta)\frac{G(\theta)}{g(\theta)}q\left(x_1\right)=0.$$

Therefore, $\Phi(\theta;\mathbf{x}_1)=0$ for all $\theta\in[\underline{\theta},\theta_H]$. Moreover, we define $\theta_H\in[\underline{\theta},\bar{\theta}]$ as the solution to

$$\int_{\theta_H}^{\bar{\theta}}\left[\frac{\partial W_1(x_1^s(\theta_H),\tilde{\theta})}{\partial x_1}g(\tilde{\theta})-(1-\delta)q\left(x_1^s(\theta_H)\right)G(\tilde{\theta})\right]d\tilde{\theta}+(1-\delta)\bar{\theta}q\left(x_1^s(\theta_H)\right)=0. \tag{32}$$

We will show at the end of this proof that this solution exists. By construction, we have

$$\Phi(\bar{\theta};\mathbf{x}_1)+(1-\delta)\bar{\theta}q\left(x_1(\bar{\theta})\right)=0,$$

and (31b) holds. Moreover, (31a) becomes

$$-\int_{\underline{\theta}}^{\bar{\theta}}\Phi(\theta;\mathbf{x}_1)d\tilde{x}_1(\theta)\leq 0\quad\forall\ \tilde{\mathbf{x}}_1\in P.$$

Assumptions 4 and $x_1^s(\theta)$ is non-increasing in θ imply that

$$\frac{\partial W_1(x_1,\theta)}{\partial x_1}\Big|_{x_1=x_1^s(\theta_H)}-(1-\delta)\frac{G(\theta)}{g(\theta)}q\left(x_1^s(\theta_H)\right)\leq 0,\quad\forall\theta\in[\theta_H,\bar{\theta}]$$

so $\Phi(\theta;\mathbf{x}_1)\leq 0$ for all $\theta\in(\theta_H,\bar{\theta}]$. Therefore, the first order condition (31) reduces to

$$\int_{\theta_H}^{\bar{\theta}}\Phi(\theta;\mathbf{x}_1)d\tilde{x}_1(\theta)\geq 0\quad\forall\ \tilde{\mathbf{x}}_1\in P,$$

which is satisfied for all non-increasing $\tilde{\mathbf{x}}_1$.

The final step is to consider the existence of a solution to equation (32). Note that

$$\frac{\partial^2 W_1(x_1^s(\theta_H),\theta)}{\partial\theta\partial x_1} = \frac{\partial}{\partial x_1}\left(\frac{\partial W_1(x_1^s(\theta_H),\theta)}{\partial\theta}\right) = \frac{\partial V_2(x_1^s(\theta_H))}{\partial x_1} = -q(x_1^s(\theta_H)).$$

Therefore, for $\theta \in [\theta_H, \bar{\theta}]$, we have

$$\frac{\partial W_1(x_1^s(\theta_H),\theta)}{\partial x_1} = \frac{\partial W_1(x_1^s(\theta_H),\theta_H)}{\partial x_1} + (\theta_H - \theta)q(x_1(\theta_H)) = (\theta_H - \theta)q(x_1(\theta_H)) + (1-\delta)\frac{G(\theta_H)}{g(\theta_H)}q\left(x_1(\theta_H)\right),$$

which allow us to simplify equation (32) in the following way

$$q(x_1^s(\theta_H))\int_{\theta_H}^{\bar{\theta}}\left[(\theta_H - \tilde{\theta}) + (1-\delta)\frac{G(\theta_H)}{g(\theta_H)} - (1-\delta)\frac{G(\tilde{\theta})}{g(\tilde{\theta})}\right]g(\tilde{\theta})d\tilde{\theta} + (1-\delta)\bar{\theta}q\left(x_1^s(\theta_H)\right) = 0.$$

Hence, we can cancel $q(x_1^s(\theta_H))$ and obtain an expression exclusively in terms of δ and $g(\theta)$

$$(1-\delta)\frac{G(\theta_H)(1-G(\theta_H))}{g(\theta_H)} + \int_{\theta_H}^{\bar{\theta}}\left[(\theta_H - \tilde{\theta})g(\tilde{\theta}) - (1-\delta)G(\tilde{\theta})\right]d\tilde{\theta} + (1-\delta)\bar{\theta} = 0.$$

Integrating by parts, we get

$$\int_{\theta_H}^{\bar{\theta}}\left[(\theta_H - \tilde{\theta})g(\tilde{\theta}) - G(\tilde{\theta})\right]d\tilde{\theta} = \theta_H - \bar{\theta}$$

so, we end with

$$H(\theta_H) \equiv (1-\delta)\frac{G(\theta_H)(1-G(\theta_H))}{g(\theta_H)} + \theta_H - \delta\left[\bar{\theta} - \int_{\theta_H}^{\bar{\theta}} G(\tilde{\theta})d\tilde{\theta}\right] = 0.$$

The function $H(\theta_H)$ satisfies the following properties

$$\begin{aligned}
H(\underline{\theta}) &= \underline{\theta} - \delta\mathbb{E}[\theta] \\
H(\bar{\theta}) &= (1-\delta)\bar{\theta} \\
H'(\theta_H) &= \left[\frac{(2-\delta G(\theta_H))}{1-G(\theta_H)}\frac{g(\theta_H)}{G(\theta_H)} - (1-\delta)\frac{g'(\theta_H)}{g(\theta_H)}\right]\frac{G(\theta_H)(1-G(\theta_H))}{g(\theta_H)}
\end{aligned}$$

Note that we can write

$$H'(\theta_H) = \left[\frac{1+\delta+G(\theta_H)}{1-G(\theta_H)} \frac{g(\theta_H)}{G(\theta_H)} + (1-\delta) \frac{\partial \log \frac{G(\theta_H)}{g(\theta_H)}}{\partial \theta_H} \right] \frac{G(\theta_H)(1-G(\theta_H))}{g(\theta_H)}$$

If $G(\theta)$ is log-concave, then $G(\theta)/g(\theta)$ is increasing, which means that $\frac{\partial \log \frac{G(\theta_H)}{g(\theta_H)}}{\partial \theta_H} > 0$. So $H'(\theta_H) > 0$. Hence, a solution to the equation (32) exists only if $\delta \geq \underline{\theta}/\mathbb{E}[\theta]$.

If $\delta < \underline{\theta}/\mathbb{E}[\theta]$, the first order condition is satisfied by setting $\theta_H = \underline{\theta}$ and $x_1(\theta) = x_1^p$ such that

$$\Phi(\bar{\theta}; \mathbf{x}_1 = x_1^p) + (1-\delta)\bar{\theta} q\left(x_1^p\right) = 0,$$

which requires that

$$\int_{\underline{\theta}}^{\bar{\theta}} \left[\frac{\partial W_1(x_1^p, \tilde{\theta})}{\partial x_1} g(\tilde{\theta}) - (1-\delta) q(x_1^p) G(\tilde{\theta}) \right] d\tilde{\theta} + (1-\delta)\bar{\theta} q(x_1^p) = 0.$$

Using integration by parts, we get that

$$x_1^p q'(x_1^p) + \left(1 - \delta \mathbb{E}[\theta]\right) q(x_1^p) = 0.$$

Noticing that $x_1^s(\underline{\theta})$ solves

$$x_1^s(\underline{\theta}) q'(x_1^s(\underline{\theta})) + (1-\underline{\theta})\, q(x_1^s(\underline{\theta})) = 0$$

we realize that whenever $\underline{\theta} > \delta \mathbb{E}[\theta]$, we have that $x_1^p > x_1^s(\underline{\theta})$. This, in turn implies that

$$\frac{\partial W_1(x_1^p, \theta)}{\partial x_1} g(\theta) - (1-\delta) q(x_1^p) G(\theta) < 0$$

so $\Phi(\theta; \mathbf{x}_1 = x_1^p) < 0$ which means that

$$-\int_{\underline{\theta}}^{\bar{\theta}} \Phi(\theta; \mathbf{x}_1 = x_1^p) d\tilde{x}_1(\theta) \leq 0 \quad \forall\ \tilde{\mathbf{x}}_1 \in P,$$

which means that $x_1(\theta) = x_1^p$ is optimal. □

Proof of Proposition 3

Suppose that $\delta = 1$ (or equivalently $\lambda = 0$). In this case, it can be readily verified that $\theta_H = \bar{\theta}$ and $x_1^s(\theta) = x_1^*(\theta)$. This candidate solution – together with the transfer at $t = 1$ defined by (10) –

solves the single lender equilibrium under incomplete information only if the financing constraint is satisfied, that is

$$\tau_0 = \mathbb{E}\left[W_1\left(x_1^*(\theta),\theta\right) + \frac{G(\theta)}{g(\theta)}V_2\left(x_1^*(\theta)\right)\right] - \bar{\theta}V_2\left(x_1^*(\bar{\theta})\right) = \bar{I}^e \geq I$$

Suppose this is not the case, and assume that

$$\mathbb{E}\left[W_1\left(x_1^*(\theta),\theta\right) + V_2\left(x_1^*(\theta)\right)\frac{G(\theta)}{g(\theta)}\right] - \bar{\theta}V_2\left(x_1^*(\bar{\theta})\right) < I,$$

We show that there exists $\delta \in [0,1]$ such that

$$\mathcal{T}_0(\delta) = \mathbb{E}\left[W_1\left(x_1^s(\theta,\delta),\theta\right) + V_2\left(x_1^s(\theta,\delta)\right)\frac{G(\theta)}{g(\theta)}\right] - \bar{\theta}V_2\left(x_1^s(\bar{\theta},\delta)\right) = I.$$

First, notice that when $\delta = 1$, we have that $x_1^s(\theta,\delta) = x_1^*(\theta)$. Hence, from our initial hypothesis, $\mathcal{T}_0(1) < I$. Next, consider the case when $\delta = 0$. In this case, we have that $\theta_H = \underline{\theta}$ and $x_1^s(\theta,\delta) = x_1^p$ where

$$x_1^p q'(x_1^p) + q(x_1^p) = 0,$$

which means that $x_1^p = \arg\max_x xq(x)$. It follows then that

$$\mathcal{T}_0(0) = x_1^p q(x_1^p) + \mathbb{E}\left[\theta + \frac{G(\theta)}{g(\theta)}\right]V_2\left(x_1^p\right) - \bar{\theta}V_2\left(x_1^p\right) = x_1^p q(x_1^p).$$

By Assumption 2, this means that $\mathcal{T}_0(0) > I$. By the maximum theorem, $x_1^s(\theta,\delta)$ is continuous in δ, and so it is $\mathcal{T}_0(\delta)$. This means that there exists some $\delta^{s*} \in [0,1]$ such that $\mathcal{T}_0(\delta^{s*}) = I$.

B Equilibrium under non-Exclusive Lending

Let us first rewrite the problem in (20). Substituting $W_1(x_1,\theta) = x_0 q(x_1) + V_1\left(x_1, x_0, \theta\right)$ together with equation (19) into the objective function in (20) and applying integration by parts,

we obtain the following problem:

$$
\begin{aligned}
W_0(x_0) = \max_{x_1(\theta)\geq x_0} \; & V_1(x_1(\bar{\theta}), x_0, \bar{\theta}) + \mathbb{E}\left[x_0 q(x_1(\theta)) - \frac{G(\theta)}{g(\theta)} V_2(x_1(\theta))\right] \\
& \text{subject to} \\
& V_1(x_1(\theta), x_0, \theta) = V_1(x_1(\bar{\theta}), x_0, \bar{\theta}) - \int_{\theta}^{\bar{\theta}} V_2(x_1(\tilde{\theta}))d\tilde{\theta} \quad \forall \theta \in \Theta \\
& x_0 \mathbb{E}\left[q(x_1(\theta))\right] \geq I \\
& x_1(\theta) \text{ is non-increasing.}
\end{aligned}
\tag{33}
$$

Proof of Lemma 2

Proof. Recall that

$$V_1(x_1, x_0, \theta) = W_1(x_1, \theta) - x_0 q(x_1).$$

For part (1), note that, by Lemma 1,

$$\frac{\partial^2 V_1(x_1, x_0, \theta)}{\partial x_1 \partial \theta} = V_2'(x_1) = -q(x_1) < 0.$$

Thus V_1 has decreasing differences in (x_1, θ). Since V_1 is concave in x_1, the monotone-comparative-statics argument of Milgrom and Shannon (1994) implies that $x_1^f(\theta, x_0)$ is nonincreasing in θ. Similarly,

$$\frac{\partial^2 V_1(x_1, x_0, \theta)}{\partial x_1 \partial x_0} = -q'(x_1) > 0,$$

so V_1 has increasing differences in (x_1, x_0), which implies that $x_1^f(\theta, x_0)$ is nondecreasing in x_0.

For part (2), $x_1^*(\theta)$ maximizes $W_1(x_1, \theta)$, so

$$\frac{\partial W_1(x_1^*(\theta), \theta)}{\partial x_1} = 0.$$

Evaluating the derivative of V_1 at $x_1 = x_1^*(\theta)$ yields

$$\frac{\partial V_1(x_1^*(\theta), x_0, \theta)}{\partial x_1} = -x_0 q'(x_1^*(\theta)) \geq 0.$$

Since V_1 is concave in x_1, its maximizer lies at or to the right of $x_1^*(\theta)$, so $x_1^f(\theta, x_0) \geq x_1^*(\theta)$.

For part (3), evaluating the derivative of V_1 at $x_1 = x_0$ gives

$$\frac{\partial V_1(x_0, x_0, \theta)}{\partial x_1} = (1-\theta)q(x_0) \geq 0,$$

where the expression follows from direct differentiation. Again by concavity, the maximizer lies at or to the right of x_0, so $x_1^f(\theta, x_0) \geq x_0$. If $\bar{\theta} = 1$, then

$$\frac{\partial V_1(x_0, x_0, \bar{\theta})}{\partial x_1} = 0.$$

Thus x_0 satisfies the first-order condition, and concavity implies $x_1^f(\bar{\theta}, x_0) = x_0$. □

Proof of Proposition 4

Proof. Letting λ be the multiplier of the investment constraint, we write

$$\max_{\theta_L} \int_{\underline{\theta}}^{\theta_L} \Big(W_1(x_1^f(\theta_L), \theta) + \lambda x_0 q(x_1^f(\theta_L))\Big)\, dG(\theta) + \int_{\theta_L}^{\bar{\theta}} \Big(W_1(x_1^f(\theta), \theta) + \lambda x_0 q(x_1^f(\theta))\Big)\, dG(\theta).$$

Dividing by $1+\lambda$ and defining $\delta \equiv 1/(1+\lambda)$, we can rewrite the previous optimization problem in the following equivalent form

$$\max_{\theta_L} \int_{\underline{\theta}}^{\theta_L} \Big((1-\delta)x_0 q(x_1^f(\theta_L)) + \delta W_1(x_1^f(\theta_L), \theta)\Big)\, dG(\theta) + \int_{\theta_L}^{\bar{\theta}} \Big((1-\delta)x_0 q(x_1^f(\theta)) + \delta W_1(x_1^f(\theta), \theta))\Big)\, dG(\theta).$$

Next, we consider the first-order condition for θ_L, which is given by

$$\begin{aligned}\frac{\partial \text{ObjFun}}{\partial \theta_L} &= x_1^{f\prime}(\theta_L) \int_{\underline{\theta}}^{\theta_L} \Big(x_0 q'(x_1^f(\theta_L)) + \delta(\theta_L - \theta) q(x_1^f(\theta_L))\Big)\, dG(\theta) \\ &= x_1^{f\prime}(\theta_L) G(\theta_L) q(x_1^f(\theta_L)) \left\{ x_0 \frac{q'(x_1^f(\theta_L))}{q(x_1^f(\theta_L))} + \delta \mathbb{E}[\theta_L - \theta | \theta \leq \theta_L] \right\} = 0\end{aligned}$$

Assumptions 3 and 1 imply that $G(\theta)$ and $q(x)$ are log-concave; accordingly, the functions $E[\theta_L - \theta | \theta \leq \theta_L]$ and $\frac{q'(x_1^f(\theta_L))}{q(x_1^f(\theta_L))}$ are increasing in θ_L. It follows that the objective function is quasi-concave, so any local maximum is also a global one. The second-order condition, evaluated at a local

maximum θ_L, is

$$\frac{\partial^2 \text{ObjFun}}{\partial \theta_L^2} = x_1^{f\prime}(\theta_L)\left[\delta G(\theta_L) + x_0 \frac{q'(x_1^f(\theta_L))}{q(x_1^f(\theta_L))} g(\theta_L)\right] q(x_1^f(\theta_L)) + \\ + \left(x_1^{f\prime}(\theta_L)\right)^2 G(\theta_L)\left[x_0 q''(x_1^f(\theta_L)) + \delta \mathbb{E}[\theta_L - \theta | \theta \leq \theta_L] q'(x_1^f(\theta_L))\right].$$

First, we verify that $\theta_L > \underline{\theta}$. Evaluating the second order condition at $\theta_L = \underline{\theta}$, we get that $\frac{\partial \text{ObjFun}}{\partial \theta_L} = 0$ and $\frac{\partial^2 \text{ObjFun}}{\partial \theta_L^2} > 0$, which means that $\theta_L = \underline{\theta}$ is a local minimum. Second, the solution is interior (that is, $\theta_L < \bar{\theta}$) only if $\left.\frac{\partial \text{ObjFun}}{\partial \theta_L}\right|_{\theta_L = \bar{\theta}} \leq 0$. As $x_1^{f'}(\theta_L) < 0$, this is the case only if

$$x_0 q'(x_1^f(\bar{\theta})) + \delta \mathbb{E}\left[\bar{\theta} - \theta\right] q(x_1^f(\bar{\theta})) \geq 0.$$

In this case, θ_L is given by the solution to

$$x_0 \frac{q'(x_1^f(\theta_L))}{q(x_1^f(\theta_L))} + \delta \mathbb{E}[\theta_L - \theta | \theta \leq \theta_L] = 0. \tag{34}$$

On the other hand, if

$$x_0 q'(x_1^f(\bar{\theta})) + \delta \mathbb{E}\left[\bar{\theta} - \theta\right] q(x_1^f(\bar{\theta})) < 0,$$

then we get that $\left.\frac{\partial \text{ObjFun}}{\partial \theta_L}\right|_{\theta_L = \bar{\theta}} > 0$, so $\theta_L = \bar{\theta} = 1$.

The next step is to consider the financing constraint. If the constraint is slack, then $\delta = 1$. Let θ_L^{uc} the solution when $\delta = 1$. The solution is interior only if

$$x_0 q'(x_1^f(\bar{\theta})) + \left(\bar{\theta} - \mathbb{E}[\theta]\right) q(x_1^f(\bar{\theta})) \geq 0.$$

Noting that Lemma 2 establishes that $x_1^f(\bar{\theta}) = x_0$ when $\bar{\theta} = 1$, this is equivalent to saying that

$$\left.\frac{\partial W_1(x_1, \mathbb{E}[\theta])}{\partial x_1}\right|_{x_1 = x_0} \geq 0,$$

which means that $x_1^*(\mathbb{E}[\theta]) \geq x_0$. If this is the case, the solution from equation (34) becomes

$$x_0 \frac{q'(x_1^f(\theta_L^{\text{uc}}))}{q(x_1^f(\theta_L^{\text{uc}}))} + \mathbb{E}[\theta_L^{\text{uc}} - \theta | \theta \leq \theta_L^{\text{uc}}] = 0$$

On the other hand, if the constraint is binding. Then, $\theta_L = \theta_L^{\text{bc}}$, where θ_L^{bc} satisfies

$$x_0 \mathbb{E}\left[q\big(x_1^f(\theta_L^{\text{bc}})\big)\,\mathbf{1}_{\{\theta \le \theta_L^{\text{bc}}\}} + q\big(x_1^f(\theta)\big)\,\mathbf{1}_{\{\theta > \theta_L^{\text{bc}}\}}\right] = I,$$

and from equation (34), we get that

$$\delta = -\frac{x_0}{\mathbb{E}[\theta_L^{\text{bc}} - \theta | \theta \le \theta_L^{\text{bc}}]} \frac{q'(x_1^f(\theta_L^{\text{bc}}))}{q(x_1^f(\theta_L^{\text{bc}}))}.$$

Finally, note that for any $\theta_L > \theta_L^{\text{bc}}$, Lemma 2 establishes that $x_1^f(\theta_L, x_0) \le x_1^f(\theta_L^{\text{bc}}, x_0)$, so, because q is decreasing,

$$x_0 \mathbb{E}\left[q\big(x_1^f(\theta_L)\big)\,\mathbf{1}_{\{\theta \le \theta_L\}} + q\big(x_1^f(\theta)\big)\,\mathbf{1}_{\{\theta > \theta_L\}}\right] > I,$$

we conclude that the solution to the optimization problem is $\max\{\theta_L^{\text{bc}}, \theta_L^{\text{uc}}\}$.

□

Proof of Proposition 5

Proof. Our verification analysis follows the approach in Amador et al. (2006) and Amador and Bagwell (2013). In particular, we find a Lagrange multiplier such that the debt limit and the multiplier are a saddle-point of the Lagrangian. The optimality of the policy then follows from the sufficient conditions in Theorem 2 in (Luenberger, 1969, p. 221). First, we will consider a Lagrange relaxation of the optimization problem in (33). Letting λ be the Lagrange multiplier of the financial constraint, we can consider the problem

$$\begin{aligned}
&\max_{x_1(\theta)} \; V_1(x_1(\bar{\theta}), x_0, \bar{\theta}) + \mathbb{E}\left[(1+\lambda)x_0 q(x_1(\theta)) - \frac{G(\theta)}{g(\theta)} V_2(x_1(\tilde{\theta}))\right] - \lambda I \\
&\text{subject to} \\
&V_1(x_1(\theta), x_0, \theta) = V_1(x_1(\bar{\theta}), x_0, \bar{\theta}) - \int_{\theta}^{\bar{\theta}} V_2(x_1(\tilde{\theta}))d\tilde{\theta} \quad \forall \theta \in \Theta \\
&x_1(\theta) \ge x_0 \\
&x_1(\theta) \text{ is non-increasing.}
\end{aligned} \tag{35}$$

Dividing by $1/(1+\lambda)$ and defining $\delta \equiv 1/(1+\lambda)$, we consider the problem

$$\begin{aligned}
&\max_{x_1(\theta)} \ \delta V_1(x_1(\bar{\theta}), x_0, \bar{\theta}) + \mathbb{E}\left[x_0 q(x_1(\theta)) - \delta \frac{G(\theta)}{g(\theta)} V_2(x_1(\tilde{\theta}))\right] \\
&\text{subject to} \\
&V_1(x_1(\theta), x_0, \theta) = V_1(x_1(\bar{\theta}), x_0, \bar{\theta}) - \int_{\theta}^{\bar{\theta}} V_2(x_1(\tilde{\theta})) d\tilde{\theta} \quad \forall \theta \in \Theta \\
&x_1(\theta) \geq x_0 \\
&x_1(\theta) \text{ is non-increasing.}
\end{aligned} \tag{36}$$

If we let $\Lambda(\theta)$ be the cumulative Lagrange multiplier for the constraint

$$V_1(x_1(\theta), x_0, \theta) = V_1(x_1(\bar{\theta}), x_0, \bar{\theta}) - \int_{\theta}^{\bar{\theta}} V_2(x_1(\tilde{\theta})) d\tilde{\theta} \quad \forall \theta \in \Theta,$$

then, the Lagrangian for our problem is

$$\begin{aligned}
\mathcal{L}(\mathbf{x}_1, \mathbf{\Lambda}) = V_1(x_1(\bar{\theta}), x_0, \bar{\theta}) \left(\delta + \Lambda(\underline{\theta}) - \Lambda(\bar{\theta})\right) + \int_{\underline{\theta}}^{\bar{\theta}} \left[x_0 q\left(x_1(\theta)\right) g(\theta) - \delta G(\theta) V_2(x_1(\theta))\right] d\theta \\
+ \int_{\underline{\theta}}^{\bar{\theta}} \left(V_1(x_1(\theta), x_0, \theta) + \int_{\theta}^{\bar{\theta}} V_2\left(x_1(\tilde{\theta})\right) d\tilde{\theta}\right) d\Lambda(\theta)
\end{aligned}$$

If we change the order of integration, we obtain that

$$\int_{\underline{\theta}}^{\bar{\theta}} \int_{\theta}^{\bar{\theta}} V_2\left(x_1(\tilde{\theta})\right) d\tilde{\theta} d\Lambda(\theta) = \int_{\underline{\theta}}^{\bar{\theta}} \Lambda(\theta) V_2(x_1(\theta)) d\theta.$$

Substituting in $\mathcal{L}(\mathbf{x}_1, \mathbf{\Lambda})$ we get the following expression for the Lagrangean.

$$\begin{aligned}
\mathcal{L}(\mathbf{x}_1, \mathbf{\Lambda}) = V_1(x_1(\bar{\theta}), x_0, \bar{\theta}) \left(\delta + \Lambda(\underline{\theta}) - \Lambda(\bar{\theta})\right) + \int_{\underline{\theta}}^{\bar{\theta}} \left[x_0 q\left(x_1(\theta)\right) g(\theta) + \left(\Lambda(\theta) - \delta G(\theta)\right) V_2(x_1(\theta))\right] d\theta \\
+ \int_{\underline{\theta}}^{\bar{\theta}} V_1(x_1(\theta), x_0, \theta) d\Lambda(\theta)
\end{aligned} \tag{37}$$

Next, we consider the following optimization problem

$$\max_{x_1(\theta)} \ \mathcal{L}(\mathbf{x}_1, \mathbf{\Lambda})$$
$$x_1(\theta) \geq x_0$$
$$x_1(\theta) \text{ is non-increasing.}$$

Denoting the partial derivative with respect to x_1 by $V_1'(x_1, x_0, \theta) \equiv \frac{\partial V_1(x_1, x_0, \theta)}{\partial x_1}$, we get that the directional derivative of $\mathcal{L}(\mathbf{x}_1, \mathbf{\Lambda})$ in the direction $\mathbf{h}$ is:

$$\begin{aligned}\nabla\mathcal{L}(\mathbf{x}_1, \mathbf{\Lambda}; \mathbf{h}) &= V_1'(x_1(\bar{\theta}), x_0, \bar{\theta}) \left(\delta + \Lambda(\underline{\theta}) - \Lambda(\bar{\theta})\right) h(\bar{\theta}) \\ &+ \int_{\underline{\theta}}^{\bar{\theta}} \left[x_0 q'\left(x_1(\theta)\right) g(\theta) - (\Lambda(\theta) - \delta G(\theta)) q(x_1(\theta))\right] h(\theta) d\theta + \int_{\underline{\theta}}^{\bar{\theta}} V_1'(x_1(\theta), x_0, \theta) h(\theta) d\Lambda(\theta) \quad (38)\end{aligned}$$

Evaluating the gradient in equation (38) at

$$x_1^l(\theta) \equiv \begin{cases} x_1^f(\theta) & \text{if } \theta \geq \theta_L \\ x_1^f(\theta_L) & \text{if } \theta < \theta_L, \end{cases}$$

we get

$$\begin{aligned}\nabla\mathcal{L}(\mathbf{x}_1^l, \mathbf{\Lambda}; \mathbf{h}) &= \int_{\theta_L}^{\bar{\theta}} \left[x_0 q'(x_1^f(\theta)) g(\theta) - (\Lambda(\theta) - \delta G(\theta)) q(x_1^f(\theta))\right] h(\theta) d\theta \\ &+ \int_{\underline{\theta}}^{\theta_L} \left[x_0 q'(x_1^f(\theta_L)) g(\theta) - (\Lambda(\theta) - \delta G(\theta)) q(x_1^f(\theta_L))\right] h(\theta) d\theta + \int_{\underline{\theta}}^{\theta_L} V_1'(x_1^f(\theta_L), x_0, \theta) h(\theta) d\Lambda(\theta)\end{aligned}$$

Let

$$\Lambda(\theta) = \begin{cases} \delta & \text{if } \theta = \bar{\theta} \\ \delta G(\theta) + \frac{x_0 q'(x_1^f(\theta))}{q(x_1^f(\theta))} g(\theta) & \text{if } \theta \in [\theta_L, \bar{\theta}) \\ 0 & \text{if } \theta \in [\underline{\theta}, \theta_L). \end{cases} \quad (39)$$

Note that from the FOC of x_1^f, we can get $V_1'(x_1^f(\theta_L), x_0, \theta) = (\theta_L - \theta) q(x_1^f(\theta_L))$ and

$$\nabla\mathcal{L}(\mathbf{x}_1^l, \mathbf{\Lambda}; \mathbf{h}) = \int_{\underline{\theta}}^{\theta_L} \left[x_0 q'(x_1^f(\theta_L)) g(\theta) + \delta G(\theta) q(x_1^f(\theta_L))\right] h(\theta) d\theta$$

We want to rewrite the directional derivative $\nabla\mathcal{L}(\mathbf{x}_1^l, \mathbf{\Lambda}; \mathbf{h})$ in terms of the increments of $\mathbf{h}(\theta)$, so we can incorporate the monotonicity constraint directly the first order conditions. With this objective in mind, we let

$$\begin{aligned}\Phi(\theta) &\equiv \int_{\underline{\theta}}^{\theta} \left[x_0 q'(x_1^f(\theta_L))g(\tilde{\theta}) + \delta G(\tilde{\theta})q(x_1^f(\theta_L))\right] d\tilde{\theta} \\ &= x_0 q'(x_1^f(\theta_L))G(\theta) + \delta q(x_1^f(\theta_L)) \int_{\underline{\theta}}^{\theta} G(\tilde{\theta})d\tilde{\theta}\end{aligned}$$

and use integration by parts to rewrite the directional derivative as

$$\nabla\mathcal{L}(\mathbf{x}_1^l, \mathbf{\Lambda}; \mathbf{h}) = \Phi(\theta_L)h(\theta_L) - \int_{\underline{\theta}}^{\theta_L} \Phi(\theta)dh(\theta).$$

Evaluating $\Phi(\theta)$ at θ_L we get

$$\Phi(\theta_L) = x_0 q'(x_1^f(\theta_L))G(\theta_L) + \delta q(x_1^f(\theta_L)) \int_{\underline{\theta}}^{\theta} G(\tilde{\theta})d\tilde{\theta} = G(\theta_L)q(x_1^f(\theta_L)) \left[x_0 \frac{q'(x_1^f(\theta_L))}{q(x_1^f(\theta_L))} + \delta\mathbb{E}[\theta_L - \theta | \theta \leq \theta_L]\right]$$

This expression is equal to zero when θ_L and δ are given by Proposition 4. Hence, we can write the directional derivative as

$$\nabla\mathcal{L}(\mathbf{x}_1^l, \mathbf{\Lambda}; \mathbf{h}) = -\int_{\underline{\theta}}^{\theta_L} \Phi(\theta)dh(\theta).$$

If $\mathcal{L}(\mathbf{x}_1, \mathbf{\Lambda})$ is concave (we will return to verify this latter), and we let P be set of all non-increasing functions on Θ, then Lemma 1 in (Luenberger, 1969, p. 227) provides the following necessary and sufficient condition for $\mathbf{x}_1^l$ to maximize $\mathcal{L}(\mathbf{x}_1, \mathbf{\Lambda})$:

$$\begin{aligned}&\nabla\mathcal{L}(\mathbf{x}_1^l, \mathbf{\Lambda}; \tilde{\mathbf{x}}_1) \leq 0 \quad \forall\ \tilde{\mathbf{x}}_1 \in P \\ &\nabla\mathcal{L}(\mathbf{x}_1^l, \mathbf{\Lambda}; \mathbf{x}_1^l) = 0,\end{aligned}$$

As $d\tilde{x}_1(\theta) \leq 0$, we need that $\Phi(\theta) \leq 0$, which is equivalent to requiring that

$$x_0 \frac{q'(x_1^f(\theta_L))}{q(x_1^f(\theta_L))} + \delta \int_{\underline{\theta}}^{\theta} \frac{G(\tilde{\theta})}{G(\theta)} d\tilde{\theta} \leq 0,\ \forall\theta \in [\underline{\theta}, \theta_L). \tag{40}$$

Using integration by parts, we get

$$\int_{\underline{\theta}}^{\theta} \frac{G(\tilde{\theta})}{G(\theta)} d\tilde{\theta} = \theta - E[\tilde{\theta}|\tilde{\theta} \leq \theta]$$

so can write equation (40) as

$$0 \geq \frac{x_0 q'(x_1^f(\theta_L))}{q(x_1^f(\theta_L))} + \delta E[\theta - \tilde{\theta}|\tilde{\theta} \leq \theta], \quad \forall \theta \in [\underline{\theta}, \theta_L).$$

Substituting $\frac{x_0 q'(x_1^f(\theta_L))}{q(x_1^f(\theta_L))}$ from Proposition 4, we get

$$1 \geq \frac{\mathbb{E}[\tilde{\theta}|\tilde{\theta} \leq \theta_L] - \mathbb{E}[\tilde{\theta}|\tilde{\theta} \leq \theta]}{\theta_L - \theta}, \quad \forall \theta \in [\underline{\theta}, \theta_L).$$

As $G(x)$ log-concave (Assumption 1), the function $\theta - \mathbb{E}[\tilde{\theta}|\tilde{\theta} \leq \theta]$ is increasing in θ (Bagnoli and Bergstrom, 2005) which means that the previous inequality is satisfied for all $\theta < \theta_L$.

It remains to verify the concavity of the $\mathcal{L}(\mathbf{x}_1, \mathbf{\Lambda})$. Substituting the $\mathbf{\Lambda}$ from equation (39) in the Lagrangian in (37), we get

$$\begin{aligned}\mathcal{L}(\mathbf{x}_1, \mathbf{\Lambda}) = &\int_{\underline{\theta}}^{\theta_L} \left[x_0 q\left(x_1(\theta)\right) + \frac{x_0 q'(x_1^f(\theta))}{q(x_1^f(\theta))} V_2(x_1(\theta)) \right] g(\theta) d\theta \\ &+ \int_{\theta_L}^{\bar{\theta}} \left[x_0 q\left(x_1(\theta)\right) - \delta \frac{G(\theta)}{g(\theta)} V_2(x_1(\theta)) \right] g(\theta) d\theta + \int_{\underline{\theta}}^{\bar{\theta}} V_1(x_1(\theta), x_0, \theta) d\Lambda(\theta) \quad (41)\end{aligned}$$

Assumption 3 implies that $q(x_1)$ is concave, so the functions $x_0 q\left(x_1\right) + \frac{x_0 q'(x_1^f(\theta))}{q(x_1^f(\theta))} V_2(x_1)$, $x_0 q\left(x_1\right) - \rho\delta \frac{G(\theta)}{g(\theta)} V_2(x_1)$ and $V_1(x_1, x_0, \theta)$ are concave in x_1 for all $\theta \in \Theta$. Hence, for the Lagrangian to be concave, it suffices that $\Lambda(\theta)$ is non-decreasing. This is the case if:

1. On $(\theta_L, \bar{\theta}]$ the function

$$\delta G(\theta) + \frac{x_0 q'(x_1^f(\theta))}{q(x_1^f(\theta))} g(\theta)$$

   is increasing.

2. At $\bar{\theta}$, $\Lambda(\theta_L) \geq \Lambda(\theta_L -)$, which requires that

$$\delta G(\theta_L) + \frac{x_0 q'(x_1^f(\theta))}{q(x_1^f(\theta))} g(\theta) \geq 0$$

For the second condition, notice that, after substituting the first order condition for θ_L in Proposition 4, we can write the condition for $\Lambda(\theta_L) \geq \Lambda(\theta_L-)$ as

$$1 - E[\theta_L - \theta | \theta \leq \theta_L] \frac{g(\theta_L)}{G(\theta_L)} = \frac{\partial}{\partial \theta_L} \left[\frac{\int_{\underline{\theta}}^{\theta_L} G(\theta) d\theta}{G(\theta_L)} \right] \geq 0,$$

which holds whenever $G(\theta)$ is log-concave. □

Proof of Proposition 6

We consider the optimization problem (28). First, notice because $x_0 q(x_0)$ is concave (from Assumption 3) and $\max_{x_0} x_0 q(x_0) \geq I$ (from Assumption 2), we know that that there are two roots $x_0^{\min} < x_0^{\max}$ to the equation $x_0 q(x_0) = I$ such that $x_0 q(x_0) \geq I$ only if $x_0 \in [x_0^{\min}, x_0^{\max}]$. Hence, we can write problem (28) as

$$\max_{x_0 \in [x_0^{\min}, x_0^{\max}]} W_0(x_0). \tag{42}$$

Let $\theta \vee \theta_L \equiv \max\{\theta, \theta_L\}$. Since Lemma 2 establishes that $x_1^f(\theta, x_0)$ is nonincreasing in θ, we can write $\min\{x_1^f(\theta), x_1^f(\theta_L)\} = x_1^f(\theta \vee \theta_L)$. Given Proposition 5, we can write $W_0(x_0)$ as

$$\begin{aligned} W_0(x_0) = & \max_{\theta_L \in \Theta} \ \mathbb{E}\left[W_1\left(x_1^f(\theta \vee \theta_L), \theta \right) \right] \\ & \text{subject to} \\ & x_0 \mathbb{E}\left[q\left(x_1^f(\theta \vee \theta_L) \right) \right] \geq I \end{aligned} \tag{43}$$

Let $W_1'(x_1, \theta) \equiv \frac{\partial W_1(x_1, \theta)}{\partial x_1}$ and let $x_1^L = x_1^f(\theta_L, x_0)$ be the debt limit in Proposition 4. By the envelope theorem, we have that for any $x_0 \in [x_0^{\min}, x_0^{\max}]$ the derivative $W_0'(x_0)$ is

$$\begin{aligned} W_0'(x_0) = \mathbb{E}\left[W_1'(x_1^f(\theta \vee \theta_L, x_0), \theta) \frac{\partial x_1^f(\theta \vee \theta_L, x_0)}{\partial x_0} \right] \\ + \lambda \mathbb{E}\left[q(x_1^f(\theta \vee \theta_L, x_0)) + x_0 q'(x_1^f(\theta \vee \theta_L, x_0)) \frac{\partial x_1^f(\theta \vee \theta_L, x_0)}{\partial x_0} \right], \end{aligned} \tag{44}$$

where $\lambda = \delta^{-1} - 1$ is the multiplier of the financing constraint $x_0 \mathbb{E}\left[q\left(x_1^f(\theta \vee \theta_L)\right)\right] \geq I$, and

$$\frac{\partial x_1^f(\theta, x_0)}{\partial x_0} = \begin{cases} \frac{q'(x_1^f(\theta,x_0))}{V_1''(x_1^f(\theta,x_0),x_0,\theta)} & \text{if } \theta < \bar{\theta} = 1 \\ 1 & \text{if } \theta = \bar{\theta} = 1. \end{cases}$$

For $\theta > \theta_L$, we have that

$$W_1'(x_1^f(\theta, x_0), \theta) = x_0 q'(x_1^f(\theta, x_0)) + V_1'(x_1^f(\theta, x_0), x_0, \theta) = x_0 q'(x_1^f(\theta, x_0));$$

while for $\theta < \theta_L$ we have

$$W_1'(x_1^f(\theta_L, x_0), \theta) = x_0 q'(x_1^f(\theta_L, x_0)) + V_1'(x_1^f(\theta_L, x_0), x_0, \theta) = x_0 q'(x_1^f(\theta_L, x_0)) + (\theta_L - \theta) q(x_1^f(\theta_L, x_0))$$

Hence, we can write the equation for $W_0'(x_0)$ as

$$\begin{aligned} W_0'(x_0) = \mathbb{E}\left[(\theta_L - \theta)\mathbf{1}_{\theta \leq \theta_L}\right] q(x_1^f(\theta_L, x_0)) \frac{\partial x_1^f(\theta_L, x_0)}{\partial x_0} \\ + \mathbb{E}\left[\lambda q(x_1^f(\theta \vee \theta_L, x_0)) + (1+\lambda) x_0 q'(x_1^f(\theta \vee \theta_L, x_0)) \frac{\partial x_1^f(\theta \vee \theta_L, x_0)}{\partial x_0}\right]. \end{aligned}$$

Noting that $\mathbb{E}[(\theta_L - \theta)\mathbf{1}_{\{\theta < \theta_L\}}] = G(\theta_L)\mathbb{E}[\theta_L - \theta | \theta \leq \theta_L]$, and letting $\lambda = \delta^{-1} - 1$, we can write

$$\begin{aligned} W_0'(x_0) = \delta^{-1} G(\theta_L) \left\{\delta \mathbb{E}[\theta_L - \theta | \theta \leq \theta_L] q(x_1^f(\theta_L, x_0)) + x_0 q'(x_1^f(\theta_L, x_0))\right\} \frac{\partial x_1^f(\theta_L, x_0)}{\partial x_0} \\ G(\theta_L)(\delta^{-1} - 1) q(x_1^f(\theta_L, x_0)) + \mathbb{E}\left[(\delta^{-1} - 1) q(x_1^f(\theta, x_0)) \mathbf{1}_{\theta > \theta_L} + \delta^{-1} x_0 q'(x_1^f(\theta, x_0)) \frac{\partial x_1^f(\theta, x_0)}{\partial x_0} \mathbf{1}_{\theta > \theta_L}\right]. \end{aligned}$$

From Proposition 4, we get that

$$\delta \mathbb{E}[\theta_L - \theta | \theta \leq \theta_L] q(x_1^f(\theta_L, x_0)) + x_0 q'(x_1^f(\theta_L, x_0)) = 0,$$

so we get that

$$W_0'(x_0) = \delta^{-1} \left\{(1-\delta) q(x_1^f(\theta_L, x_0)) G(\theta_L) + \mathbb{E}\left[(1-\delta) q(x_1^f(\theta, x_0)) \mathbf{1}_{\theta > \theta_L} + x_0 q'(x_1^f(\theta, x_0)) \frac{\partial x_1^f(\theta, x_0)}{\partial x_0} \mathbf{1}_{\theta > \theta_L}\right]\right\}.$$

Hence, we can write $W_0^{'}(x_0)$ as

$$W_0^{'}(x_0) = \delta^{-1}\mathbb{E}\left[(1-\delta)q(x_1^f(\theta \vee \theta_L, x_0)) + x_0 q'(x_1^f(\theta, x_0))\frac{\partial x_1^f(\theta, x_0)}{\partial x_0}\mathbf{1}_{\theta>\theta_L}\right]. \tag{45}$$

If the solution is interior, that is $x_0^* > x_0^{\min}$, then we have that $W_0^{'}(x_0) = 0$, in which case we get the following necessary first order

$$x_0 = \frac{(1-\delta)\mathbb{E}\left[q(x_1^f(\theta_L, x_0))\mathbf{1}_{\{\theta\leq\theta_L\}} + q(x_1^f(\theta, x_0))\mathbf{1}_{\{\theta>\theta_L\}}\right]}{-\mathbb{E}\left[q'(x_1^f(\theta, x_0))\frac{\partial x_1^f(\theta,x_0)}{\partial x_0}\mathbf{1}_{\{\theta>\theta_L\}}\right]}.$$

We can conclude then that if this is the case, then (x_0, θ_L, δ) satisfy

$$\begin{aligned}
x_0 &= \frac{(1-\delta)\mathbb{E}\left[q(x_1^f(\theta_L, x_0))\mathbf{1}_{\{\theta\leq\theta_L\}} + q(x_1^f(\theta, x_0))\mathbf{1}_{\{\theta>\theta_L\}}\right]}{-\mathbb{E}\left[q'(x_1^f(\theta, x_0))\frac{\partial x_1^f(\theta,x_0)}{\partial x_0}\mathbf{1}_{\{\theta>\theta_L\}}\right]} \\
\delta &= -\frac{x_0}{\mathbb{E}[\theta_L - \theta|\theta \leq \theta_L]}\frac{q'(x_1^f(\theta_L))}{q(x_1^f(\theta_L))} \\
x_0\mathbb{E}&\left[q(x_1^f(\theta_L, x_0))\mathbf{1}_{\{\theta\leq\theta_L\}} + q(x_1^f(\theta, x_0))\mathbf{1}_{\{\theta>\theta_L\}}\right] = I.
\end{aligned}$$

It remains to determine under what conditions the solution is interior. We proceed through a series of lemmas characterizing the solution to the optimization problem (42)

Lemma 4. *Let x_0^* be a solution to problem (42) and θ_L^* the associated solution to problem (43). If the financing constraint $x_0^*\mathbb{E}\left[q\left(x_1^f(\theta \vee \theta_L, x_0^*)\right)\right] \geq I$ is slack, then $x_0 \geq x_1^*(\mathbb{E}[\theta])$.*

Proof. We proceed by contradiction. As $x_1^L = x_1^f(\theta_L, x_0) > x_0$, and Lemma 2 establishes that $x_1^f(\bar{\theta}, x_0) = x_0$, it must be the case that $x_0 \in (x_0^{\min}, x_0^{\max})$ because $x_0 \in \{x_0^{\min}, x_0^{\max}\}$ does not satisfy the constraint $x_0\mathbb{E}\left[q\left(\min\{x_1^f(\theta), x_1^f(\theta_L)\}\right)\right] \geq I$. This means, that x_0^* is in the interior $[x_0^{\min}, x_0^{\max}]$, so it must satisfy the necessary condition $W_0^{'}(x_0) = 0$. This means that $\lambda = 0$ and $\theta_L < \bar{\theta}$, so the expression in equation (44) reduces to

$$W_0^{'}(x_0) = \mathbb{E}\left[W_1'(x_1^f(\theta \vee \theta_L, x_0), \theta)\frac{\partial x_1^f(\theta \vee \theta_L, x_0)}{\partial x_0}\right].$$

From here we get that

$$
\begin{aligned}
W_0^{'}(x_0) &= \mathbb{E}\left[W_1'(x_1^f(\theta \vee \theta_L, x_0), \theta)\frac{\partial x_1^f(\theta \vee \theta_L, x_0)}{\partial x_0}\right] \\
&= \mathbb{P}(\theta > \theta_L)\mathbb{E}\left[W_1'(x_1^f(\theta, x_0), \theta)\frac{\partial x_1^f(\theta, x_0)}{\partial x_0}\Big|\theta > \theta_L\right] \\
&\quad + \mathbb{P}(\theta \leq \theta_L)\mathbb{E}\left[W_1'(x_1^f(\theta_L, x_0), \theta)\Big|\theta \leq \theta_L\right]\frac{\partial x_1^f(\theta_L, x_0)}{\partial x_0}.
\end{aligned}
$$

Suppose that $x_0 < x_1^*(\mathbb{E}[\theta])$. If the constraint is slack then equation (26) implies that

$$
\mathbb{E}\left[W_1'(x_1^f(\theta_L, x_0), \theta)\Big|\theta \leq \theta_L\right] = 0.
$$

Moreover, Lemma 2 establishes that $x_1^f(\theta, x_0) \geq x_1^*(\theta)$, so $W_1'(x_1^f(\theta, x_0), \theta) \leq 0$; on the relevant interior branch the inequality is strict. It follows then that

$$
W_0^{'}(x_0) = \mathbb{P}(\theta > \theta_L)\mathbb{E}\left[W_1'(x_1^f(\theta, x_0), \theta)\frac{\partial x_1^f(\theta, x_0)}{\partial x_0}\Big|\theta > \theta_L\right] < 0,
$$

which yields a contradiction.

□

From here, we can conclude that if the financing constraint is slack. then it must be the case that $x_0 \geq x_1^*(\mathbb{E}[\theta])$. The next lemma shows that in any solution, the constraint must be binding.

Lemma 5. *If x_0^* is a solution to problem (42) and θ_L^* the associated solution to problem (43). Then, the financing constraint is binding, i.e., $x_0^*\mathbb{E}\left[q\left(x_1^f(\theta \vee \theta_L, x_0^*)\right)\right] = I$.*

Proof. We proceed by contradiction. Suppose that the constraint is slack. Then, it must be the case that $x_0^* \in (x_0^{\min}, x_0^{\max})$, and by Lemma 4 it must be the case that $x_0^* \geq x_1^*(\mathbb{E}[\theta])$. From Proposition 4, we get that $\theta_L^* = \bar{\theta}$ and $\lambda = 0$. Next, we show that x_0 cannot achieve a local maximum at x_0^*. As $W_1(x_1, \mathbb{E}[\theta])$ is concave in x_1, and achieves a maximum at $x_1^*(\mathbb{E}[\theta])$, we have that we have $W_0^{'}(x_0) < 0$ for any $x_0 > x_1^*(\mathbb{E}[\theta])$, so $x_0^* > x_1^*(\mathbb{E}[\theta])$ cannot be optimal. Hence, if $x_0^* \geq x_1^*(\mathbb{E}[\theta])$ it must be equal to $x_1^*(\mathbb{E}[\theta])$. If $x_0^* = x_1^*(\mathbb{E}[\theta]) \in (x_0^{\min}, x_0^{\max})$, then the constraint must be slack for $x_0 = x_1^*(\mathbb{E}[\theta])$. By continuity, it is also slack for $x_0' = x_1^*(\mathbb{E}[\theta]) - \epsilon$ (for some $\epsilon > 0$), which means that

$$
W_0^{'}(x_0') = \mathbb{E}\left[W_1'(x_1^f(\theta \vee \theta_L, x_0'), \theta)\frac{\partial x_1^f(\theta \vee \theta_L, x_0')}{\partial x_0}\right] < 0.
$$

Taking the limit when $\epsilon \to 0$, we get that $W_0'(x_1^*(\mathbb{E}[\theta])-) < 0$, which means that $x_1^*(\mathbb{E}[\theta])$ cannot be optimal. We conclude that the constraint cannot be slack at the optimal solution.

□

The final step is to characterize the conditions under which $x_0^* > x_0^{\min}$.

Lemma 6. *Let x_0^* be a solution to the problem (42). If $x_0^{\min} \geq x_1^*(\mathbb{E}[\theta])$; then $x_0^* = x_0^{\min}$. If $x_0^{\min} < x_1^*(\mathbb{E}[\theta])$ then $x_0^* > x_0^{\min}$.*

Proof. We consider two cases: $x_0^{\min} \geq x_1^*(\mathbb{E}[\theta])$ and $x_0^{\min} < x_1^*(\mathbb{E}[\theta])$. If $x_0^{\min} \geq x_1^*(\mathbb{E}[\theta])$, then Proposition 4 implies that for any $x_0 \in [x_0^{\min}, x_0^{\max}]$, we have $\theta_L = \bar{\theta} = 1$, which means that $x_0^* = \arg\max_{x_0 \in [x_0^{\min}, x_0^{\max}]} W_1(x_0, \mathbb{E}[\theta])$. As $W_1(x_0, \mathbb{E}[\theta])$ is a concave function (by Assumption 3), achieving its maximum at $x_1^*(\mathbb{E}[\theta]) \leq x_0^{\min}$. It follows that $x_0^* = x_0^{\min}$. On the other hand, if $x_0^{\min} \geq x_1^*(\mathbb{E}[\theta])$, then $W_1(x_0, \mathbb{E}[\theta])$ is increasing at $x_0 = x_0^{\min}$, which means that $x_0^* > x_0^{\min}$ □

Proof of Proposition 7

The proof works directly on the binding frontier. First, we show that the optimal solution satisfies $\hat{x}_0 \in [0, 1/2]$, so on the relevant branch the binding constraint gives a unique frontier choice $\hat{x}_0 = \chi(\theta_L; \phi, \kappa)$ for each feasible θ_L. Second, Proposition 6 reduces the interior binding optimum to a one-dimensional equation in θ_L. Comparative statics then follow from the monotonicity of the one-dimensional equation.

For any function $h(\theta)$ and $\phi \in [\underline{\phi}, \overline{\phi}]$ let

$$\mathbb{E}_\phi[h(\theta)] \equiv \int_{\underline{\theta}}^1 h(\theta) g(\theta; \phi)\, d\theta,$$

and let

$$\hat{U}(\hat{x}_0, \theta_L; \phi) = \int_{\underline{\theta}}^1 \hat{u}(\hat{x}_0, \theta_L, \theta) g(\theta; \phi)\, d\theta,$$
$$\hat{C}(\hat{x}_0, \theta_L; \phi) = \hat{x}_0(1 - \hat{x}_0) K(\theta_L; \phi)$$

with

$$K(\theta_L; \phi) \equiv \mathbb{E}_\phi \left[\frac{1}{2 - \theta_L} \mathbf{1}_{\{\theta \leq \theta_L\}} + \frac{1}{2 - \theta} \mathbf{1}_{\{\theta > \theta_L\}} \right] = \int_{\underline{\theta}}^1 \frac{g(\theta; \phi)}{2 - \max\{\theta, \theta_L\}}\, d\theta,$$

and

$$\hat{u}(\hat{x}_0, \theta_L, \theta) = \begin{cases} \dfrac{1-\hat{x}_0}{2-\theta_L} - \dfrac{(1-\theta/2)(1-\hat{x}_0)^2}{(2-\theta_L)^2}, & \theta \le \theta_L, \\ \dfrac{1-\hat{x}_0^2}{2(2-\theta)}, & \theta > \theta_L. \end{cases}$$

The borrower problem at time zero in equation (28) can be written as

$$\max_{(\hat{x}_0,\theta_L)\in[0,1]\times\Theta} \hat{U}(\hat{x}_0, \theta_L; \phi) \quad \text{s.t.} \quad \hat{C}(\hat{x}_0, \theta_L; \phi) \ge \kappa.$$

Lemma 7. *For any feasible pair $(\hat{x}_0, \theta_L)$ with $\hat{x}_0 > 1/2$, the mirrored pair $(1-\hat{x}_0, \theta_L)$ is feasible and yields weakly higher objective. Hence an optimizer exists in*

$$\mathcal{X}^- \equiv [0, 1/2] \times \Theta.$$

Proof. Let $\hat{x}_0' > 1/2$ and $\hat{x}_0^- \equiv 1 - \hat{x}_0' \in [0, 1/2)$. Since

$$\hat{x}_0^-(1-\hat{x}_0^-) = \hat{x}_0'(1-\hat{x}_0'),$$

we have $\hat{C}(\hat{x}_0^-, \theta_L; \phi) = \hat{C}(\hat{x}_0', \theta_L; \phi)$, so feasibility is unchanged. Now compare payoffs pointwise. For $\theta > \theta_L$:

$$\hat{u}(\hat{x}_0^-, \theta_L, \theta) - \hat{u}(\hat{x}_0', \theta_L, \theta) = \frac{2\hat{x}_0' - 1}{2(2-\theta)} > 0.$$

For $\theta \le \theta_L$:

$$\hat{u}(\hat{x}_0^-, \theta_L, \theta) - \hat{u}(\hat{x}_0', \theta_L, \theta) = (2\hat{x}_0' - 1)\frac{1-\theta_L+\theta/2}{(2-\theta_L)^2} \ge 0.$$

Taking expectation with respect to θ, we get

$$\hat{U}(\hat{x}_0^-, \theta_L; \phi) \ge \hat{U}(\hat{x}_0', \theta_L; \phi).$$

□

Lemma 8. *Fix (κ, ϕ) and restrict attention to $\mathcal{X}^- = [0, 1/2]$. Define*

$$\begin{aligned} \Theta(\phi, \kappa) &\equiv \{\theta_L \in \Theta : K(\theta_L; \phi) \ge 4\kappa\} \\ \chi(\theta_L; \phi, \kappa) &\equiv \frac{1 - \sqrt{1 - \frac{4\kappa}{K(\theta_L;\phi)}}}{2} \\ V(\theta_L; \phi, \kappa) &\equiv \hat{U}(\chi(\theta_L; \phi, \kappa), \theta_L; \phi). \end{aligned}$$

Then:

1. $K_\theta(\theta_L;\phi) = G(\theta_L;\phi)/(2-\theta_L)^2 > 0$, *so* $\Theta(\phi,\kappa)$ *is an interval.*

2. *If* $(\hat{x}_0, \theta_L) \in \mathcal{X}^-$ *is binding, then* $\theta_L \in \Theta(\phi,\kappa)$ *and*

$$\hat{x}_0 = \chi(\theta_L;\phi,\kappa).$$

3. *If the optimum on* $\mathcal{X}^-$ *is binding and unique, then its second coordinate is the unique maximizer of* $V(\cdot;\phi,\kappa)$ *on* $\Theta(\phi,\kappa)$.

Proof. Using

$$K(\theta_L;\phi) = \int_{\underline{\theta}}^{\theta_L} \frac{g(\theta;\phi)}{2-\theta_L}\,d\theta + \int_{\theta_L}^{1} \frac{g(\theta;\phi)}{2-\theta}\,d\theta,$$

Leibniz' rule gives

$$K_\theta(\theta_L;\phi) = \frac{G(\theta_L;\phi)}{(2-\theta_L)^2} > 0.$$

Hence $K(\cdot;\phi)$ is strictly increasing, so $\Theta(\phi,\kappa)$ is an interval. On $[0,1/2]$, the map $x \mapsto x(1-x)$ is strictly increasing. Therefore, the binding equation $\hat{x}_0(1-\hat{x}_0)K(\theta_L;\phi) = \kappa$ has the unique solution $\hat{x}_0 = \chi(\theta_L;\phi,\kappa)$ on that branch. The last claim is immediate from this one-to-one parameterization of the binding frontier. □

We can reduce the system in Proposition 6 to single dimensional equation in θ_L. Define

$$\begin{aligned}
A(\theta_L;\phi) &\equiv \mathbb{E}_\phi\left[\frac{1}{2-\theta}\mathbf{1}_{\{\theta>\theta_L\}}\right] = \int_{\theta_L}^{1} \frac{g(\theta;\phi)}{2-\theta}\,d\theta \\
\rho(\theta_L;\phi) &\equiv \mathbb{E}_\phi[\theta_L - \theta \mid \theta \le \theta_L] \\
\alpha(\theta_L;\phi) &\equiv \frac{2-\theta_L}{\rho(\theta_L;\phi)} \\
\Psi(\theta_L;\phi) &\equiv \frac{K(\theta_L;\phi)}{A(\theta_L;\phi) + (1+\alpha(\theta_L;\phi))K(\theta_L;\phi)}
\end{aligned}$$

Also define

$$
\begin{aligned}
R(\theta_L;\phi,\kappa) &\equiv \chi(\theta_L;\phi,\kappa) - \Psi(\theta_L;\phi) \\
B(\theta_L;\phi) &\equiv \frac{A(\theta_L;\phi)}{K(\theta_L;\phi)} + \alpha(\theta_L;\phi) \\
\beta(\theta_L;\phi,\kappa) &\equiv \frac{1+\sqrt{1-\frac{4\kappa}{K(\theta_L;\phi)}}}{1-\sqrt{1-\frac{4\kappa}{K(\theta_L;\phi)}}}.
\end{aligned}
$$

With this notation, we can write the system of equations in Proposition 6 as

$$
\begin{aligned}
\kappa &= \hat{x}_0(1-\hat{x}_0)K(\theta_L;\phi) \\
\frac{\hat{x}_0}{1-\hat{x}_0} &= \frac{(1-\delta)K(\theta_L;\phi)}{A(\theta_L;\phi)} \\
\delta &= \alpha(\theta_L;\phi)\frac{\hat{x}_0}{1-\hat{x}_0}
\end{aligned}
\tag{46}
$$

Lemma 9 (Scalar equation for the interior binding optimum)**.** *Suppose the solution is characterized by the binding system (46). Then, on $\Theta(\phi,\kappa)$:*

1. *$(\chi(\theta_L;\phi,\kappa),\theta_L)$ solves the interior binding system if and only if*

$$R(\theta_L;\phi,\kappa) = 0.$$

2. *Equivalently,*

$$R(\theta_L;\phi,\kappa) = 0 \quad \Longleftrightarrow \quad B(\theta_L;\phi) = \beta(\theta_L;\phi,\kappa).$$

Proof. From the second and third equations in (46), with $r \equiv \hat{x}_0/(1-\hat{x}_0)$, we have

$$r = \frac{(1-\alpha r)K}{A} \quad \Longrightarrow \quad r = \frac{K}{A+\alpha K}.$$

Hence

$$\hat{x}_0 = \frac{r}{1+r} = \frac{K}{A+(1+\alpha)K} = \Psi(\theta_L;\phi).$$

The first equation in (46) gives $\hat{x}_0 = \chi(\theta_L;\phi,\kappa)$ on $\mathcal{X}^-$. Therefore the interior binding system is equivalent to $R(\theta_L;\phi,\kappa) = 0$. Next,

$$\Psi(\theta_L;\phi) = \frac{1}{1+B(\theta_L;\phi)}$$

because

$$\frac{1}{1+B} = \frac{1}{1+A/K+\alpha} = \frac{K}{A+(1+\alpha)K}.$$

Likewise,

$$\chi(\theta_L;\phi,\kappa) = \frac{1}{1+\beta(\theta_L;\phi,\kappa)}.$$

Therefore

$$R(\theta_L;\phi,\kappa) = \frac{B(\theta_L;\phi)-\beta(\theta_L;\phi,\kappa)}{(1+B(\theta_L;\phi))(1+\beta(\theta_L;\phi,\kappa))},$$

so $R(\theta_L;\phi,\kappa)=0$ if and only if $B(\theta_L;\phi)=\beta(\theta_L;\phi,\kappa)$. □

The next lemma establishes that the single-dimensional equation determining θ_L is monotonic.

Lemma 10. *Fix (κ,ϕ) and suppose $G(\cdot;\phi)$ is log-concave on Θ. Then, on the interior of $\Theta(\phi,\kappa)$,*

$$\begin{aligned}\chi_\theta(\theta_L;\phi,\kappa) &< 0\\ \Psi_\theta(\theta_L;\phi) &> 0\\ R_\theta(\theta_L;\phi,\kappa) &< 0.\end{aligned}$$

In particular, $R(\cdot;\phi,\kappa)$ has at most one zero.

Proof. Since $K_\theta > 0$,

$$\chi_\theta(\theta_L;\phi,\kappa) = -\frac{\kappa K_\theta(\theta_L;\phi)}{K(\theta_L;\phi)^2\sqrt{1-\frac{4\kappa}{K(\theta_L;\phi)}}} < 0.$$

Now let $D \equiv A+(1+\alpha)K$. Then

$$\Psi_\theta(\theta_L;\phi) = \frac{AK_\theta - KA_\theta - \alpha_\theta K^2}{D^2},$$

with

$$K_\theta(\theta_L;\phi) = \frac{G(\theta_L;\phi)}{(2-\theta_L)^2} > 0, \qquad A_\theta(\theta_L;\phi) = -\frac{g(\theta_L;\phi)}{2-\theta_L} < 0,$$

and, under log-concavity of $G(\cdot;\phi)$, we also have $\rho(\theta_L;\phi) > 0$ and $\rho_\theta(\theta_L;\phi) \geq 0$ on the interior branch, because $\rho(\cdot;\phi)$ is the mean-advantage-over-inferior function associated with $G(\cdot;\phi)$, wihch is increasing under log-concavity of $G(\cdot;\phi)$ (Bagnoli and Bergstrom, 2005, Theorem 5). It follows that

$$\alpha_\theta(\theta_L;\phi) = -\frac{1}{\rho(\theta_L;\phi)} - \frac{(2-\theta_L)\rho_\theta(\theta_L;\phi)}{\rho(\theta_L;\phi)^2} < 0.$$

Therefore $\Psi_\theta(\theta_L;\phi) > 0$, and hence

$$R_\theta(\theta_L;\phi,\kappa) = \chi_\theta(\theta_L;\phi,\kappa) - \Psi_\theta(\theta_L;\phi) < 0.$$

□

The next lemma establishes the impact of ϕ implied by the MLR order on several expectations.

Lemma 11. *Fix $\theta_L \in \Theta$ and suppose $\phi' > \phi$. Then:*

1. $K(\theta_L;\phi') \geq K(\theta_L;\phi)$.
2. $A(\theta_L;\phi')/K(\theta_L;\phi') \geq A(\theta_L;\phi)/K(\theta_L;\phi)$.
3. $\rho(\theta_L;\phi') \leq \rho(\theta_L;\phi)$, *and therefore* $\alpha(\theta_L;\phi') \geq \alpha(\theta_L;\phi)$.

Consequently,

$$B(\theta_L;\phi') \geq B(\theta_L;\phi).$$

Proof. Let

$$m_K(\theta;\theta_L) \equiv \frac{1}{2-\max\{\theta,\theta_L\}}, \qquad m_A(\theta;\theta_L) \equiv \frac{\mathbf{1}_{\{\theta\geq\theta_L\}}}{2-\theta}.$$

For the first inequality, as m_K is increasing in θ, and MLR implies first-order stochastic dominance (FOSD), it follows that

$$K(\theta_L;\phi') = \mathbb{E}_{\phi'}[m_K(\theta;\theta_L)] \geq \mathbb{E}_{\phi}[m_K(\theta;\theta_L)] = K(\theta_L;\phi).$$

For the second inequality, notice that

$$\frac{m_A(\theta;\theta_L)}{m_K(\theta;\theta_L)} = \mathbf{1}_{\{\theta\geq\theta_L\}}$$

is increasing in θ. Define the tilted density

$$q(\theta;\phi,\theta_L) \equiv \frac{m_K(\theta;\theta_L)g(\theta;\phi)}{K(\theta_L;\phi)}.$$

Then

$$\frac{A(\theta_L;\phi)}{K(\theta_L;\phi)} = \int \frac{m_A(\theta;\theta_L)}{m_K(\theta;\theta_L)} q(\theta;\phi,\theta_L)\,d\theta = \int \mathbf{1}_{\{\theta\geq\theta_L\}} q(\theta;\phi,\theta_L)\,d\theta.$$

So $A(\theta_L;\phi)/K(\theta_L;\phi)$ is the upper-tail probability of θ_L under the tilted law $q(\cdot;\phi,\theta_L)$. Since

$$\frac{q(\theta;\phi',\theta_L)}{q(\theta;\phi,\theta_L)} = \frac{K(\theta_L;\phi)}{K(\theta_L;\phi')}\,\frac{g(\theta;\phi')}{g(\theta;\phi)},$$

and the first factor is constant in θ, the MLR ordering of the family $\{g(\cdot;\phi)\}$ implies that $q(\cdot;\phi',\theta_L)$ dominates $q(\cdot;\phi,\theta_L)$ in the MLR order. Hence $q(\cdot;\phi',\theta_L)$ first-order stochastically dominates $q(\cdot;\phi,\theta_L)$, and therefore the tail probability above is larger:

$$\frac{A(\theta_L;\phi')}{K(\theta_L;\phi')} \geq \frac{A(\theta_L;\phi)}{K(\theta_L;\phi)}.$$

Finally, MLR implies reverse-hazard order (Shaked and Shanthikumar, 2007, Theorem 1.C.1). Equivalently,

$$\theta \mapsto \frac{G(\theta;\phi')}{G(\theta;\phi)}$$

is increasing. Therefore, for every $\tilde{\theta} \leq \theta_L$,

$$\frac{G(\tilde{\theta};\phi')}{G(\theta_L;\phi')} \leq \frac{G(\tilde{\theta};\phi)}{G(\theta_L;\phi)}.$$

Using

$$\rho(\theta_L;\phi) = \int_{\underline{\theta}}^{\theta_L} \frac{G(\tilde{\theta};\phi)}{G(\theta_L;\phi)}\, d\tilde{\theta},$$

we obtain $\rho(\theta_L;\phi') \leq \rho(\theta_L;\phi)$ and hence $\alpha(\theta_L;\phi') \geq \alpha(\theta_L;\phi)$. The claim for B follows immediately. □

Lemma 12. *Fix $\theta_L \in \Theta(\phi,\kappa)$.*

1. *If $\kappa' \geq \kappa$, then $\chi(\theta_L;\phi,\kappa') \geq \chi(\theta_L;\phi,\kappa)$.*
2. *If $\phi' > \phi$, then $\beta(\theta_L;\phi',\kappa) \leq \beta(\theta_L;\phi,\kappa)$.*

Proof. For fixed (θ_L,ϕ), $\chi(\theta_L;\phi,\kappa)$ is increasing in κ by inspection of the closed form. For the second claim, $\beta(\theta_L;\phi,\kappa)$ is a decreasing function of $K(\theta_L;\phi)$, and Lemma 11(1) gives $K(\theta_L;\phi') \geq K(\theta_L;\phi)$. □

Proof of Proposition 7: By Lemmas 7 and 8, the unique binding optimum on $\mathcal{X}^-$ is the unique zero of $R(\cdot;\phi,\kappa)$ on $\Theta(\phi,\kappa)$.

- θ_L^* *in* κ. Fix ϕ and let $\kappa' \geq \kappa$. Lemma 12(1) gives

$$R(\theta_L;\phi,\kappa') \geq R(\theta_L;\phi,\kappa) \qquad \forall \theta_L \in \Theta(\phi,\kappa'),$$

because Ψ does not depend on κ. Since $R(\cdot;\phi,\kappa')$ is strictly decreasing by Lemma 10, its unique zero must lie at or to the right of the unique zero of $R(\cdot;\phi,\kappa)$. Therefore

$$\theta_L^*(\kappa',\phi) \geq \theta_L^*(\kappa,\phi).$$

- $\hat{x}_0^*$ *in* κ. On the interior branch,

$$\hat{x}_0^*(\kappa,\phi) = \Psi(\theta_L^*(\kappa,\phi);\phi).$$

By Lemma 10, $\Psi_\theta > 0$, and we already proved that $\theta_L^*(\kappa,\phi)$ is nondecreasing in κ. Hence $\hat{x}_0^*(\kappa,\phi)$ is nondecreasing in κ.

- θ_L^* *in* ϕ. Fix κ and let $\theta_L^* := \theta_L^*(\kappa,\phi)$. By Lemma 9,

$$B(\theta_L^*;\phi) = \beta(\theta_L^*;\phi,\kappa).$$

By Lemmas 11 and 12,

$$B(\theta_L^*;\phi') \geq B(\theta_L^*;\phi) = \beta(\theta_L^*;\phi,\kappa) \geq \beta(\theta_L^*;\phi',\kappa).$$

Hence Lemma 9 gives

$$R(\theta_L^*;\phi',\kappa) \geq 0.$$

Since $R(\cdot;\phi',\kappa)$ is strictly decreasing by Lemma 10, its unique zero must satisfy

$$\theta_L^*(\kappa,\phi') \geq \theta_L^*(\kappa,\phi).$$

- $\hat{x}_0^*$ *in* ϕ. On the binding branch,

$$\hat{x}_0^*(\kappa,\phi) = \chi(\theta_L^*(\kappa,\phi);\phi,\kappa).$$

Now $\chi(\cdot;\phi,\kappa)$ is decreasing in θ_L by Lemma 10, and $\chi(\theta_L;\cdot,\kappa)$ is decreasing in ϕ because $K(\theta_L;\phi)$ is increasing in ϕ by Lemma 11(1). Since $\theta_L^*(\kappa,\phi') \geq \theta_L^*(\kappa,\phi)$, we conclude

$$\hat{x}_0^*(\kappa,\phi') \leq \hat{x}_0^*(\kappa,\phi).$$

C Extensions

C.1 Short-Term Debt

Motivated by the solution in Section 4, we consider contracts (b, x_0, x_1^L), where b is the amount of short-term debt that matures at $t = 1$, x_0 is the initial amount of long-term debt that matures at $t = 2$, and x_1^L is the limit in total debt due at $t = 2$. As we did before, it is convenient to consider the problem in terms of the marginal type θ_L for which the debt limit binds, that is, $x_1^f(\theta_L) = x_1^L$. Because the borrower is financially constrained, the following roll-over constraint must be satisfied at $t = 1$:

$$q(x_1(\theta))(x_1(\theta) - x_0) \geq b.$$

This constraint specifies that the amount raised at $t = 1$ must be sufficient to pay back the maturing short-term debt. It is possible to roll over the debt only if $q(x_1^L)(x_1^L - x_0) \geq b$. Moreover, if $\bar{\theta} = 1$ (Assumption 5), Lemma 2 establishes that $x_1^f(\bar{\theta}) - x_0 = 0$, which means that there is a marginal type θ_H such that

$$q(x_1^f(\theta_H))(x_1^f(\theta_H) - x_0) = b.$$

Hence, we can consider the following generalization of the problem in (25)

$$\begin{aligned}
&\max_{\substack{\theta_L,\theta_H\in\Theta\\ \theta_L\leq\theta_H}} \int_{\underline{\theta}}^{\theta_L} W_1(x_1^f(\theta_L),\theta)dG(\theta) + \int_{\theta_L}^{\theta_H} W_1(x_1^f(\theta),\theta)dG(\theta) + \int_{\theta_H}^{\bar{\theta}} W_1(x_1^f(\theta_H),\theta)dG(\theta) \\
&\text{subject to} \\
&x_0\left[\left(q(x_1^f(\theta_L)) - q(x_1^f(\theta_H))\right)G(\theta_L) + \int_{\theta_L}^{\theta_H}\left(q(x_1^f(\theta)) - q(x_1^f(\theta_H))\right)dG(\theta)\right] \\
&+ x_1^f(\theta_H)q(x_1^f(\theta_H)) \geq I.
\end{aligned} \tag{47}$$

Let λ be the multiplier of the investment constraint. We can write

$$\begin{aligned}
\max_{\substack{\theta_L,\theta_H\in\Theta\\ \theta_L\leq\theta_H}} &\int_{\underline{\theta}}^{\theta_L}\left(W_1(x_1^f(\theta_L),\theta) + \lambda x_0 q(x_1^f(\theta_L))\right)dG(\theta) + \int_{\theta_L}^{\theta_H}\left(W_1(x_1^f(\theta),\theta) + \lambda x_0 q(x_1^f(\theta))\right)dG(\theta) \\
&+ \int_{\theta_H}^{\bar{\theta}}\left(W_1(x_1^f(\theta_H),\theta) + \lambda x_0 q(x_1^f(\theta_H))\right)dG(\theta) + \lambda q(x_1^f(\theta_H))(x_1^f(\theta_H) - x_0).
\end{aligned}$$

Dividing by $1+\lambda$ and letting $\delta = 1/(1+\lambda)$, we get that the previous problem is equivalent to

$$\max_{\substack{\theta_L,\theta_H\in\Theta\\ \theta_L\leq\theta_H}} \int_{\underline{\theta}}^{\theta_L} \Big((1-\delta)x_0 q(x_1^f(\theta_L)) + \delta W_1(x_1^f(\theta_L),\theta)\Big)\, dG(\theta) + \int_{\theta_L}^{\theta_H} \Big((1-\delta)x_0 q(x_1^f(\theta)) + \delta W_1(x_1^f(\theta),\theta))\Big)\, dG(\theta)$$
$$+ \int_{\theta_H}^{\bar{\theta}} \Big((1-\delta)x_0 q(x_1^f(\theta_H)) + \delta W_1(x_1^f(\theta_H),\theta))\Big)\, dG(\theta) + (1-\delta)q(x_1^f(\theta_H))(x_1^f(\theta_H)-x_0).$$

If we ignore the constraint $\theta_L \leq \theta_H$, the first-order condition for θ_L is the same as in the case without short-term debt. So, we get the condition

$$x_0 \frac{q'(x_1^f(\theta_L))}{q(x_1^f(\theta_L))} + \delta\mathbb{E}[\theta_L - \theta | \theta \leq \theta_L] = 0.$$

Next, we consider the first-order condition for θ_H; which is given by

$$\frac{\partial \text{ObjFun}}{\partial \theta_H} = x_1^{f'}(\theta_H)\left\{\int_{\theta_H}^{\bar{\theta}} \Big(x_0 q'(x_1^f(\theta_H)) + \delta(\theta_H-\theta)q(x_1^f(\theta_H))\Big)\, dG(\theta) + (1-\delta)\theta_H q(x_1^f(\theta_H))\right\}$$
$$= (1-G(\theta_H))x_1^{f'}(\theta_H)q(x_1^f(\theta_H))\left\{x_0\frac{q'(x_1^f(\theta_H))}{q(x_1^f(\theta_H))} - \delta\mathbb{E}[\theta-\theta_H|\theta\geq\theta_H] + (1-\delta)\frac{\theta_H}{1-G(\theta_H)}\right\} = 0,$$

where we have used the following relationships

$$x_1^f(\theta_H)q'(x_1^f(\theta_H)) + (1-\theta)q(x_1^f(\theta_H)) = (\theta_H-\theta)q(x_1^f(\theta_H)) + x_0 q'(x_1^f(\theta))$$
$$(x_1^f(\theta_H)-x_0)q'(x_1^f(\theta_H)) + q(x_1^f(\theta_H)) = \theta_H q(x_1^f(\theta_H)).$$

From here, we get the first-order condition

$$x_0\frac{q'(x_1^f(\theta_H))}{q(x_1^f(\theta_H))} - \delta\mathbb{E}[\theta-\theta_H|\theta\geq\theta_H] + (1-\delta)\frac{\theta_H}{1-G(\theta_H)} = 0.$$

We now prove that the financing constraint at $t=0$ must bind. If not, then $\delta = 1$, and we have

$$\underbrace{x_0\frac{q'(x_1^f(\theta_H))}{q(x_1^f(\theta_H))}}_{<0} - \underbrace{\mathbb{E}[\theta-\theta_H|\theta\geq\theta_H]}_{>0} < 0.$$

In this case, $\frac{\partial \text{ObjFun}}{\partial \theta_H} > 0$, so that $\theta_H = \bar{\theta}$, which is a contradiction. Therefore, the financing constraint is binding, that is, $\delta < 1$, and $(\theta_L, \theta_H, \delta)$ satisfy

$$
\begin{aligned}
&x_0 \frac{q'(x_1^f(\theta_H))}{q(x_1^f(\theta_H))} - \delta \mathbb{E}[\theta - \theta_H | \theta \geq \theta_H] + (1-\delta)\frac{\theta_H}{1 - G(\theta_H)} = 0 \\
&x_0 \frac{q'(x_1^f(\theta_L))}{q(x_1^f(\theta_L))} + \delta \mathbb{E}[\theta_L - \theta | \theta \leq \theta_L] = 0 \\
&x_0 \left[G(\theta_L) q(x_1^f(\theta_L)) + \int_{\theta_L}^{\theta_H} q(x_1^f(\theta)) dG(\theta) - G(\theta_H) q(x_1^f(\theta_H)) \right] + x_1^f(\theta_H) q(x_1^f(\theta_H)) = I.
\end{aligned}
$$

The final step is to consider the first-order condition for x_0. Let $W_0(x_0)$ be the value function of the optimization problem in equation (47). From here, we get

$$
\begin{aligned}
W_0'(x_0) = {} & x_0 \int_{\theta_L}^{\theta_H} q'(x_1^f(\theta)) \frac{\partial x_1^f(\theta)}{\partial x_0} dG(\theta) + \frac{\partial x_1^f(\theta_L)}{\partial x_0} \int_{\underline{\theta}}^{\theta_L} \Big(x_0 q'(x_1^f(\theta_L)) + \delta(\theta_L - \theta) q(x_1^f(\theta_L)) \Big) dG(\theta) \\
& + \frac{\partial x_1^f(\theta_H)}{\partial x_0} \left\{ \int_{\theta_H}^{\bar{\theta}} \Big(x_0 q'(x_1^f(\theta_H)) + \delta(\theta_H - \theta) q(x_1^f(\theta_H)) \Big) dG(\theta) + (1-\delta)\theta_H q(x_1^f(\theta_H)) \right\} \\
& + (1-\delta) \left[\int_{\underline{\theta}}^{\theta_L} q(x_1^f(\theta_L)) dG(\theta) + \int_{\theta_L}^{\theta_H} q(x_1^f(\theta)) dG(\theta) + \int_{\theta_H}^{\bar{\theta}} q(x_1^f(\theta_H)) dG(\theta) - q(x_1^f(\theta_H)) \right] \\
= {} & x_0 \int_{\theta_L}^{\theta_H} \underbrace{q'(x_1^f(\theta)) \frac{\partial x_1^f(\theta)}{\partial x_0}}_{<0} dG(\theta) \\
& + (1-\delta) \left[\underbrace{\Big(q(x_1^f(\theta_L)) - q(x_1^f(\theta_H)) \Big)}_{<0} G(\theta_L) + \int_{\theta_L}^{\theta_H} \underbrace{\Big(q(x_1^f(\theta)) - q(x_1^f(\theta_H)) \Big)}_{<0} dG(\theta) \right] < 0,
\end{aligned}
$$

where the sign restrictions in the second line follow from Lemma 2, which establishes that $x_1^f(\cdot, x_0)$ is nonincreasing, together with the fact that q is decreasing.

From here, we get that the solution is $x_0 = 0$, so $x_1^f(\theta) = x_1^*(\theta)$. Substituting back into the first order conditions for θ_L and θ_H we get that $\theta_L = \underline{\theta}$ and

$$
\begin{aligned}
&x_1^*(\theta_H) q(x_1^*(\theta_H)) = I \\
&\delta = \frac{\theta_H}{\theta_H + \mathbb{E}[\theta - \theta_H | \theta \geq \theta_H](1 - G(\theta_H))}.
\end{aligned}
$$

The result in the proposition follows by setting

$$x_1^b = x_1^*(\theta_H).$$

C.2 Mandatory Prepayment

Proof of Proposition 9

Proof. Our main objective is to show that $x_1^+(\theta) = x^{s*}(\theta)$ and $x_0 = x^{s*}(\bar{\theta})$ solve the optimization problem (29). First, by the envelope theorem, we can write

$$m_1(\theta) = V_1(x_1^+(\theta), x_0, \theta) - \bar{V}_1 + \int_\theta^{\bar{\theta}} V_2(x_1^+(\tilde{\theta}))d\tilde{\theta},$$

where $\bar{V}_1 = V_1(x_1^+(\bar{\theta}), x_0, \bar{\theta}) - m(\bar{\theta})$. Substituting in the budget constraint and using integration by parts, we get

$$\mathbb{E}\left[x_0 q(x_1^+(\theta)) + m_1(\theta)\right] = \mathbb{E}\left[W_1(x_1^+(\theta), \theta) + V_2(x_1^+(\theta))\frac{G(\theta)}{g(\theta)}\right] - \bar{V}_1.$$

Next, using the definition of $V_1(x_1, x_0, \theta)$

$$V_1(x_1^+(\theta), x_0, \theta) = (x_1^+(\theta) - x_0)q(x_1^+(\theta)) + \theta V_2(x_1^+(\theta)),$$

together with the envelope representation for $m_1(\theta)$, we can write the constraint $(x_1^+(\theta) - x_0)q(x_1^+(\theta)) \geq m_1(\theta)$ as

$$\bar{V}_1 - \theta V_2(x_1^+(\theta)) - \int_\theta^{\bar{\theta}} V_2(x_1^+(\tilde{\theta}))d\tilde{\theta} \geq 0.$$

If $x_1^+(\theta)$ is non-increasing, then $\int_\theta^{\bar{\theta}} V_2(x_1^+(\tilde{\theta}))d\tilde{\theta} + \theta V_2(z_1(\theta))$ is increasing, so we only need to enforece the constraint $(x_1^+(\theta) - x_0)q(x_1^+(\theta)) \geq m_1(\theta)$ at $\theta = \bar{\theta}$, which yields

$$(x_1^+(\bar{\theta}) - x_0)q(x_1^+(\bar{\theta})) \geq m_1(\bar{\theta}) = V_1(x_1^+(\bar{\theta}), x_0, \bar{\theta}) - \bar{V}_1.$$

Substituting $V_1(x_1^+(\bar{\theta}), x_0, \bar{\theta})$, we obtain

$$\bar{V}_1 \geq \bar{\theta} V_2(x_1^+(\theta)).$$

Next, notice that for any $x_1^+(\theta) < x_1^f(\theta)$, we have that $\frac{V_1(x_1^+(\theta),x_0,\theta)}{\partial x_1} > 0$; hence, we have that if $x_1^+(\theta) < x_1^f(\theta)$ and $x_1^+(\theta)$ is non-increasing, then

$$m_1'(\theta) = \frac{V_1(x_1^+(\theta), x_0, \theta)}{\partial x_1} x_1^{+'}(\theta) \leq 0,$$

hence, the constraint $m_1(\theta) \geq 0$ only needs to be enforced at $\theta = \bar{\theta}$, where $m_1(\bar{\theta}) = V_1(x_1^+(\bar{\theta}), x_0, \bar{\theta}) - \bar{V}_1$. So we get constraint

$$V_1(x_1^+(\bar{\theta}), x_0, \bar{\theta}) \geq \bar{V}_1$$

Finally, we consider the constraint $m_1(\theta) \leq x_0 \left(1 - q(x_1^+(\theta))\right)$. Substituting the expression for $m_1(\theta)$ from the envelope condition, we get

$$V_1(x_1^+(\theta), x_0, \theta) - \bar{V}_1 + \int_\theta^{\bar{\theta}} V_2(x_1^+(\tilde{\theta}))d\tilde{\theta} \leq (1 - q(x_1^+(\theta)))x_0.$$

Next, we substitute

$$V_1(x_1^+(\theta), x_0, \theta) = W_1(x_1^+(\theta), \theta) - x_0 q(x_1^+(\theta))$$

to get

$$W_1(x_1^+(\theta), \theta) - \bar{V}_1 + \int_\theta^{\bar{\theta}} V_2(x_1^+(\tilde{\theta}))d\tilde{\theta} \leq x_0.$$

Let's define the function $H(\theta)$ as

$$H(\theta) \equiv W_1(x_1^+(\theta), \theta) - \bar{V}_1 + \int_\theta^{\bar{\theta}} V_2(x_1^+(\tilde{\theta}))d\tilde{\theta},$$

so the condition becomes $H(\theta) \leq 0$. Differentiating the previous expression, we get

$$H'(\theta) = W_1'(x_1^+(\theta), \theta) x_1^{+'}(\theta).$$

If $x_1^+(\theta) = x_1^{s*}(\theta)$, then we get that

$$W_1'(x_1^+(\theta), \theta) = (1 - \delta)\frac{G(\theta)}{g(\theta)} q(x_1^+) > 0$$

so $H'(\theta) < 0$ unless $x_1^+ = 0$. In particular, this is true as long as $x_1^+(\theta) < x_1^*(\theta)$. In particular, we get the single solution allocation $x_1^+(\theta) = x_1^{s*}(\theta)$, satisfies the constraint $m_1(\theta) \leq x_0 \left(1 - q(x_1^+(\theta))\right)$

for all $\theta \in \Theta$ if $H(\underline{\theta}) \leq x_0$, which means that

$$W_1(x_1^+(\underline{\theta}), \underline{\theta}) - \theta_H V_2(x_1^+(\theta_H)) + \int_{\underline{\theta}}^{\theta_H} V_2(x_1^+(\tilde{\theta}))d\tilde{\theta} \leq x_1^+(\theta_H).$$

It follows that we can consider a relaxed version of problem (29) as

$$\begin{aligned}
&\max_{x_0, x_1^+(\theta), \overline{V}_1} \mathbb{E}\left[W_1(x_1^+(\theta), \theta)\right] \\
&\text{subject to} \\
&\mathbb{E}\left[W_1(x_1^+(\theta), \theta) + V_2(x_1^+(\theta))\frac{G(\theta)}{g(\theta)}\right] - \bar{V}_1 \geq I \\
&W_1(x_1^+(\underline{\theta}), \underline{\theta}) - \bar{V}_1 + \int_{\underline{\theta}}^{\bar{\theta}} V_2(x_1^+(\tilde{\theta}))d\tilde{\theta} \leq x_0 \\
&\overline{V}_1 \in [V_1(x_1^+(\bar{\theta}), x_0, \bar{\theta}), \bar{\theta}V_2(x_1^+(\bar{\theta}))] \\
&x_1^+(\theta) \text{ is non-increasing}
\end{aligned}$$

First, notice that this problem is feasible only if $V_1(x_1^+(\bar{\theta}), x_0, \bar{\theta}) \leq \bar{\theta}V_2(x_1^+(\bar{\theta}))$ which immediately yields $x_0 = x_1^+(\bar{\theta})$ and $\overline{V}_1 = \bar{\theta}V_2(x_1^+(\bar{\theta}))$. Hence, we can write

$$\begin{aligned}
&\max_{x_1^+(\theta)} \mathbb{E}\left[W_1(x_1^+(\theta), \theta)\right] \\
&\text{subject to} \\
&\mathbb{E}\left[W_1(x_1^+(\theta), \theta) + V_2(x_1^+(\theta))\frac{G(\theta)}{g(\theta)}\right] - \bar{\theta}V_2(x_1^+(\bar{\theta})) \geq I \\
&W_1(x_1^+(\underline{\theta}), \underline{\theta}) - \bar{\theta}V_2(x_1^+(\bar{\theta})) + \int_{\underline{\theta}}^{\bar{\theta}} V_2(x_1^+(\tilde{\theta}))d\tilde{\theta} \leq x_0 \\
&x_1^+(\theta) \text{ is non-increasing}
\end{aligned}$$

If we consider a relaxed version in which we ignore the constraint $m_1(\bar{\theta}) \leq x_0\left(1 - q(x_1^+(\bar{\theta}))\right)$, and – as we did before – we define the shadow discount factor $\delta \equiv 1/(1+\lambda)$, where λ is the multiplier

of the budget constraint; then, we can write the optimization problem for given δ as

$$\max_{x_1^+(\theta), m_1(\theta)} \mathbb{E}\left[W_1(x_1^+(\theta),\theta) + (1-\delta)\frac{G(\theta)}{g(\theta)}V_2(x_1^+(\theta))\right] - (1-\delta)\bar{\theta}V_2(x_1^+(\bar{\theta}))$$

subject to

$x_1^+(\theta)$ is non-increasing,

which is identical to the problem in the single lender case. Which means that $x_1^+(\theta) = x_1^{s*}(\theta)$ as long as

$$W_1(x_1^{s*}(\underline{\theta}),\underline{\theta}) - \bar{\theta}V_2(x_1^{s*}(\bar{\theta})) + \int_{\underline{\theta}}^{\bar{\theta}} V_2(x_1^{s*}(\tilde{\theta}))d\tilde{\theta} \leq x_1^{s*}(\bar{\theta})$$

The final step then is to show that this condition is satisfied by $x_1^{s*}(\theta)$. As $x_1^{s*}(\bar{\theta}) + \bar{\theta}V_2(x_1^{s*}(\bar{\theta})) \geq W_1(x_1^{s*}(\bar{\theta}),\bar{\theta})$, it suffices to show that

$$W_1(x_1^{s*}(\underline{\theta}),\underline{\theta}) + \int_{\underline{\theta}}^{\bar{\theta}} V_2(x_1^{s*}(\tilde{\theta}))d\tilde{\theta} \leq W_1(x_1^{s*}(\bar{\theta}),\bar{\theta}).$$

Differentiating $W_1(x_1^{s*}(\theta),\theta)$ we get that

$$\begin{aligned}\frac{d}{d\theta}W_1(x_1^{s*}(\theta),\theta) &= (1-\delta)\frac{G(\theta)}{g(\theta)}q(x_1^{s*}) + V_2(x_1^{s*}(\theta)) \\ &\geq V_2(x_1^{s*}(\theta)),\end{aligned}$$

so it follows from here that

$$W_1(x_1^{s*}(\bar{\theta}),\bar{\theta}) \geq W_1(x_1^{s*}(\underline{\theta}),\underline{\theta}) + \int_{\underline{\theta}}^{\bar{\theta}} V_2(x_1^{s*}(\tilde{\theta}))d\tilde{\theta},$$

whch immediately implies the required inequality.

□

C.2.1 Proof Proposition 10

Most of the argument for the implementation of the mechanism is standard. The only part that is not standard is the verification that $x_1^s(\theta) + b(\theta)$ is invertible on $[\underline{\theta}, \theta_H]$.

$$\tilde{x}_1(\theta) = x_1^s(\theta) + b(\theta) = x_1^s(\theta) + \frac{m_1(\theta)}{1 - q(x_1^s(\theta))}$$

Differentiating, we get

$$
\begin{aligned}
\tilde{x}_1'(\theta) &= x_1^{s'}(\theta) + \frac{x_1^{s'}(\theta)}{1-q(x_1^s(\theta)}\left[V_1'(x_1^s(\theta), x_0, \theta) + q'(x_1^s(\theta))\frac{m_1(\theta)}{1-q(x_1^s(\theta))}\right] \\
&= \frac{x_1^{s'}(\theta)}{1-q(x_1^s(\theta)}\left[1-q(x_1^s(\theta)) + V_1'(x_1^s(\theta), x_0, \theta) + q'(x_1^s(\theta))\left(\tilde{x}_1(\theta) - x_1^s(\theta)\right)\right] \\
&= \frac{x_1^{s'}(\theta)}{1-q(x_1^s(\theta)}\left[1-q(x_1^s(\theta)) + (x_1^s(\theta)-x_0)q'(x_1^s(\theta)) + (1-\theta)q(x_1^s(\theta)) + q'(x_1^s(\theta))\left(\tilde{x}_1(\theta) - x_1^s(\theta)\right)\right] \\
&= \frac{x_1^{s'}(\theta)}{1-q(x_1^s(\theta)}\left[1-\theta q(x_1^s(\theta)) + q'(x_1^s(\theta))\left(\tilde{x}_1(\theta) - x_0\right)\right] \\
&= \frac{x_1^{s'}(\theta)}{1-q(x_1^s(\theta)}\left[1-\theta q(x_1^s(\theta)) + q'(x_1^s(\theta))\left(x_1^s(\theta) + b(\theta) - x_0\right)\right]
\end{aligned}
$$

As $b(\theta) \in [0, x_0]$ and $q'(x_1) < 0$ we have that

$$
\begin{aligned}
1-\theta q(x_1^s(\theta)) + q'(x_1^s(\theta))\left(x_1^s(\theta) + b(\theta) - x_0\right) &\geq 1-\theta q(x_1^s(\theta)) + q'(x_1^s(\theta))\left(x_1^s(\theta) + x_0 - x_0\right) \\
&= 1-\theta q(x_1^s(\theta)) + q'(x_1^s(\theta))x_1^s(\theta) \\
&\geq (1-\theta)q(x_1^s(\theta)) + q'(x_1^s(\theta))x_1^s(\theta) \\
&= W_1'(x_1^s(\theta), \theta) > 0.
\end{aligned}
$$

It follows that $\operatorname{sign} \tilde{x}_1'(\theta) = \operatorname{sign} x_1^{s'}(\theta) \leq 0$, with strict inequality when $\theta < \theta_H$. Hence, we can define the inverse $\theta(x_1) = x_1^{-1}(x_1)$ and we can write the prepayment as a function of the issuance Δx_1 at $t = 1$ as $b(\theta(x_0 + \Delta x_1)$. The rest of the implementation corresponds to specifying $b(\theta(x_0 + \Delta x_1)$ as a share from the proceeds from new issuance. In particular, defining $\alpha(\Delta x_1)$ as the solution to the equation $b(\theta(x_0 + \Delta x_1) = \alpha(\Delta x_1)q(x_0 + \Delta x_1 - b(\theta(x_0 + \Delta x_1))\Delta x_1$ we can specify a non-linear debt sweep

$$
\alpha(\Delta x_1) = \frac{b(\theta(x_0 + \Delta x_1)}{q\left(x_0 + \Delta x_1 - b(\theta(x_0 + \Delta x_1)\right)\Delta x_1}.
$$